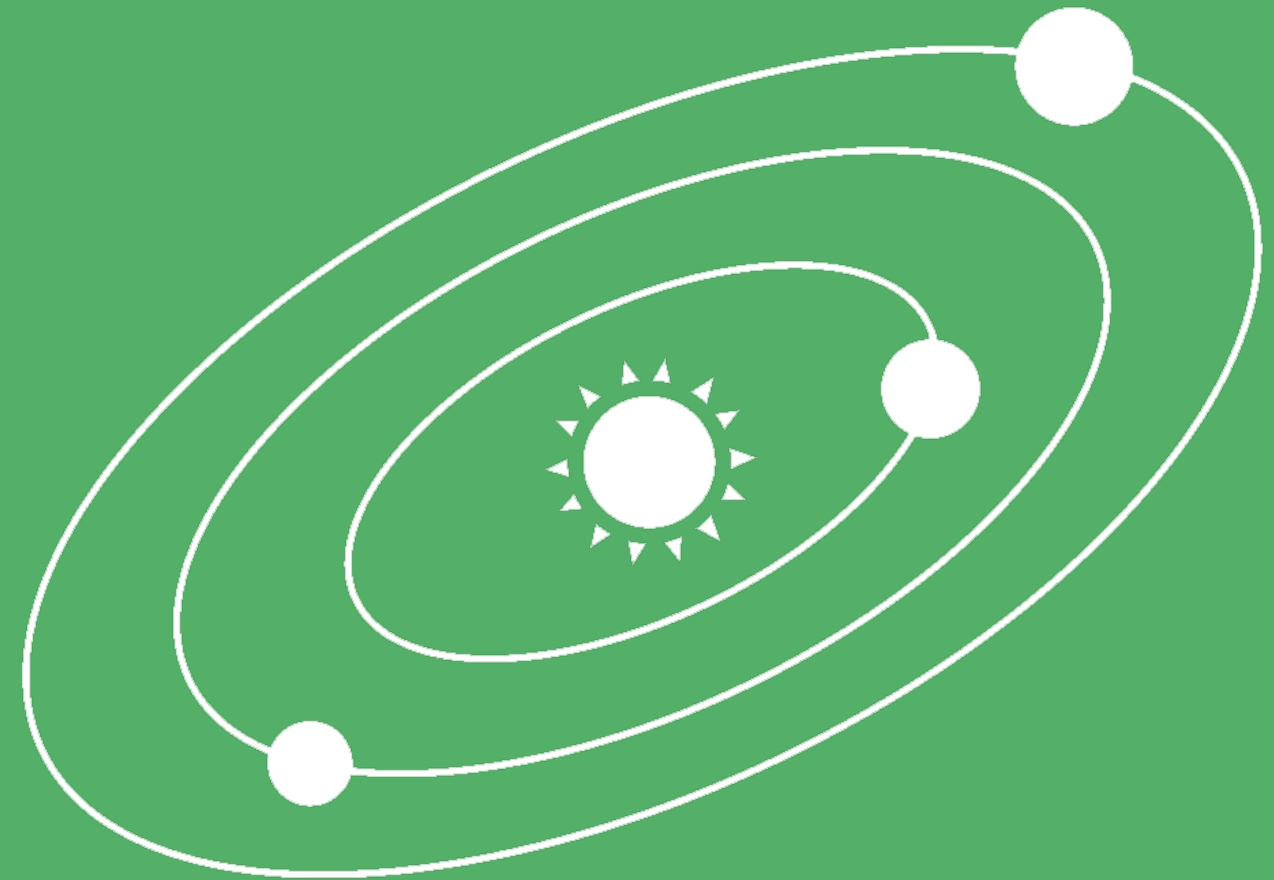


Systems Modeling

A Field Guide



What we cover in this guide

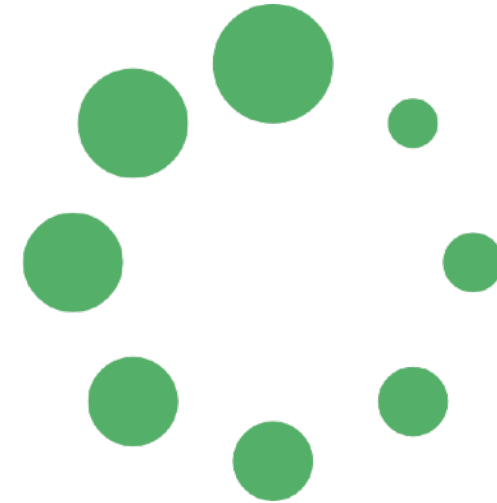
This guide is designed to introduce you to systems modeling. The guide presents a lightweight modeling language that builds upon standard concepts within systems theory, such as; efficiency, energy and entropy, integration, emergence, functions, environments, etc. What is presented here is primarily qualitative in nature, designed to provide a generic language for structuring our reasoning about systems of all kinds.

The primary aim of systems modeling is to be able to shift our thinking from that of seeing just parts to seeing whole systems and to be able to communicate this insight. During this guide, readers will learn the language of systems theory and learn how to graphically represent systems. These diagrams and vocabulary will enable us to reason about a wide variety of phenomena while being able to effectively communicate this to others.



1. Systems Models

Set & Systems
Elements
Relations
Emergence



2. Functions

Systems Function
Energy & Entropy
Efficiency & Dissipation
Inertia



3. Environment

Systems Boundary
Autonomy
Environment
Hierarchy



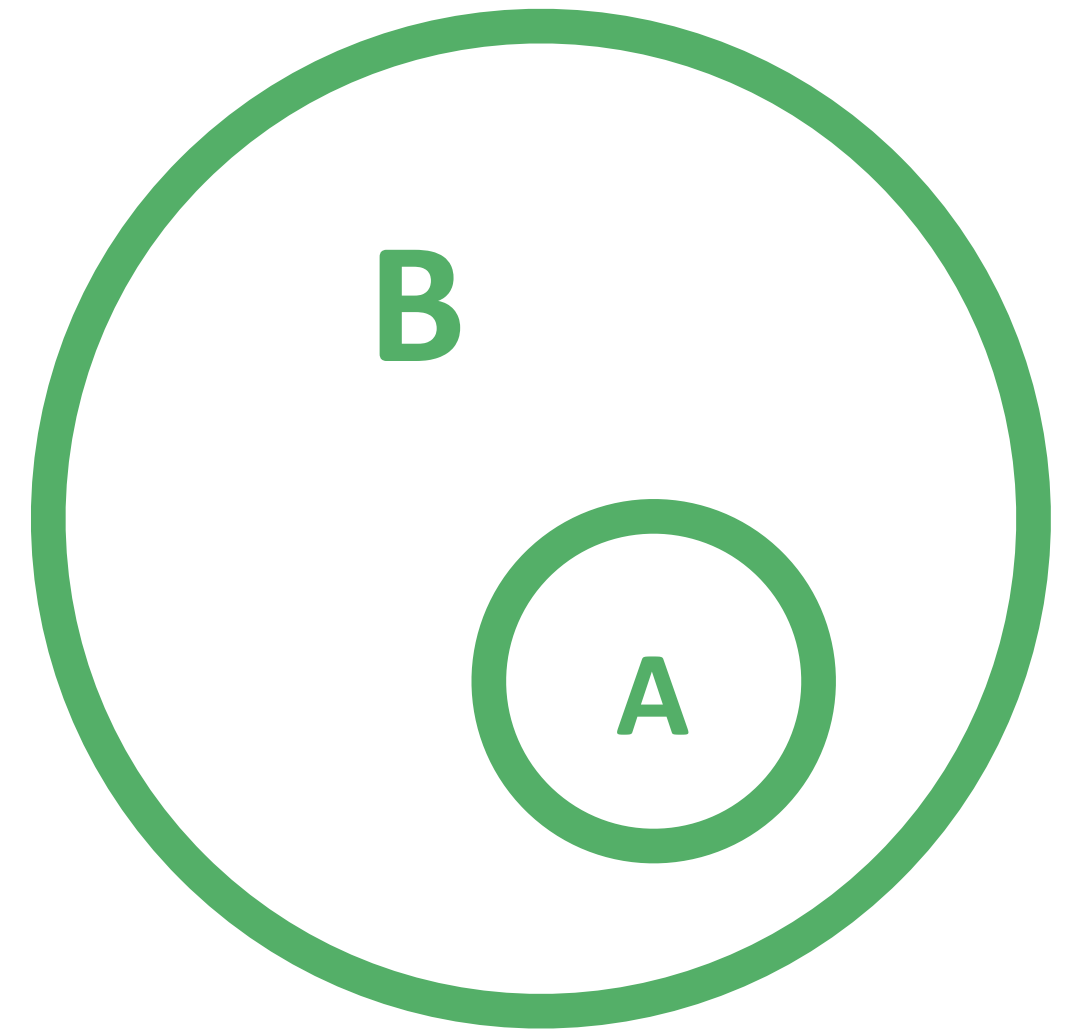
4. Dynamics

Feedback
System Dynamics
Regulatory Systems
Homeostasis

Why Models?

Most systems in the real world are too complex for us to grasp as a whole due to their many diverse components and interactions. In practice, we can typically only interact with and experience a small subset of a system. As such it is models that, in some way, actually **enable us to grasp the whole system**, but only ever through the use of abstraction that creates a simplified representation of the actual real-world phenomenon.

A model is an abstract and compact representation of some phenomenon that **enables us to conceptualize and communicate its basic structure and dynamics in a coherent fashion**. All models are simplified representations of reality. Models require the compression of information or data to generate a simplified form. This simplification process helps us to represent the whole of a system, to organize and structure information, clarify our reasoning, communicate, solve problems and predict events.



Abstraction

Models are created through the use of abstraction. Abstraction is the process of **removing various levels of detail from the original empirical phenomenon** in order to create a compact schema or diagram of the system under consideration.

Abstraction helps us to synthesize our many experiences of some entity into a coherent impression of the whole. For example, one may have a model as to what a nation like Brazil is like, but one could only ever, in reality, interact with a very limited subset of the different elements that constitute it.

The use of abstraction involves removing successive layers of detail from a representation in order to **capture only the essential features that are generic to all entities of that kind** and independent of their specific form. Examples of abstractions include; an algebraic equation, a map of a city, an architect's master plan of a building, or a non-figurative painting.



Perspectives

To be effective, a model must capture diverse information, perspectives and views on a particular phenomenon. For example, if we take some complex entity like a city, there will be multiple perspectives on how to interpret it, social, technological, economic, demographic, etc. An effective model must be able to integrate all of these diverse perspectives in some way, to give us a vision of the whole system and a basic understanding of its primary constituent parts and relations.

Models are not facts they are structured and coherent ways of looking at the world thus there is no such thing as a model that is correct or incorrect. They are simply more or less useful to us depending on what they enable us to see vs what they inhibit us from seeing. An effective model is a balancing act between simplicity, through abstraction, and breadth or scope in order to include all of the different views and possible instances of the system.



Effectiveness



The effectiveness of a model may be defined along a number of different parameters. For example, **how well does the model allow us to grasp the whole and identify its core attributes?** Does it manage to synthesize all of the different perspectives on it? How faithful is it to the empirical phenomena? Does it stand up to empirical testing? Moreover, can it predict future events?

All models contain assumptions and making them explicit is important to effective modeling. Models only ever represent some subset of all possible phenomena and thus have to make certain assumptions about other elements and systems outside of our focus of interest. **Effective models make explicit the assumptions that they entail.**

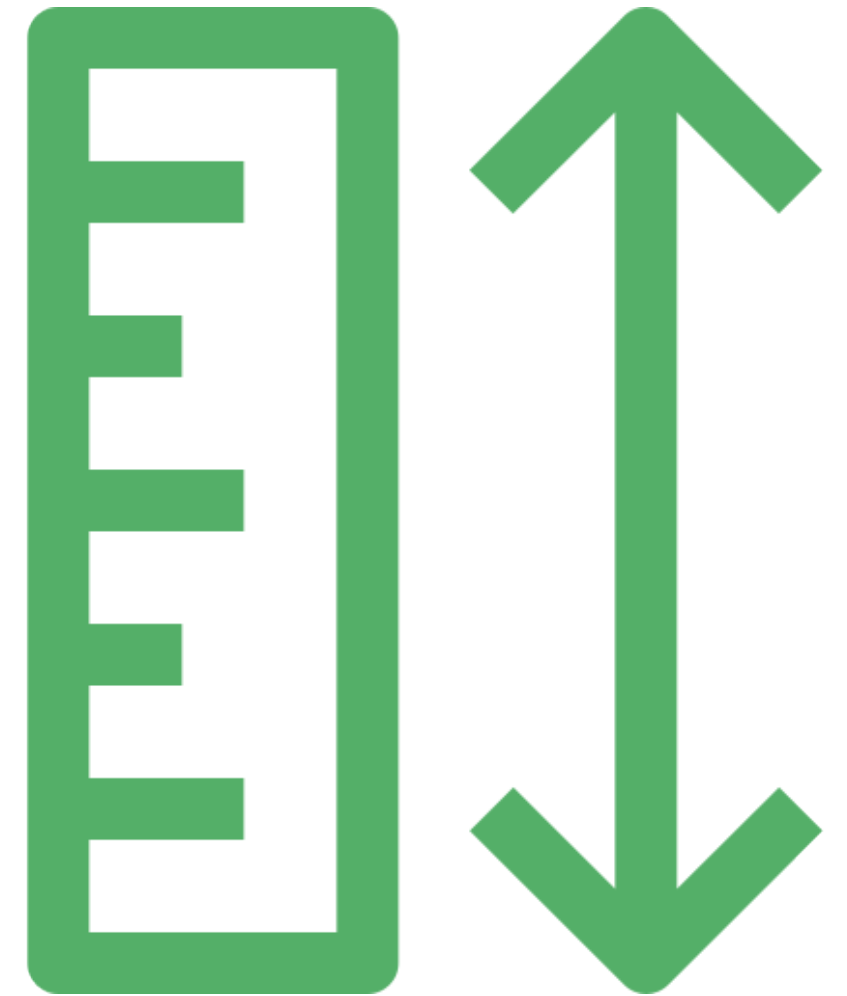
We should be asking how solid are the model's foundational assumptions? That is to say, the assumptions that it is based upon. Are they truly self-evident solid assumptions or are they contingent upon certain conditions that may not always hold?

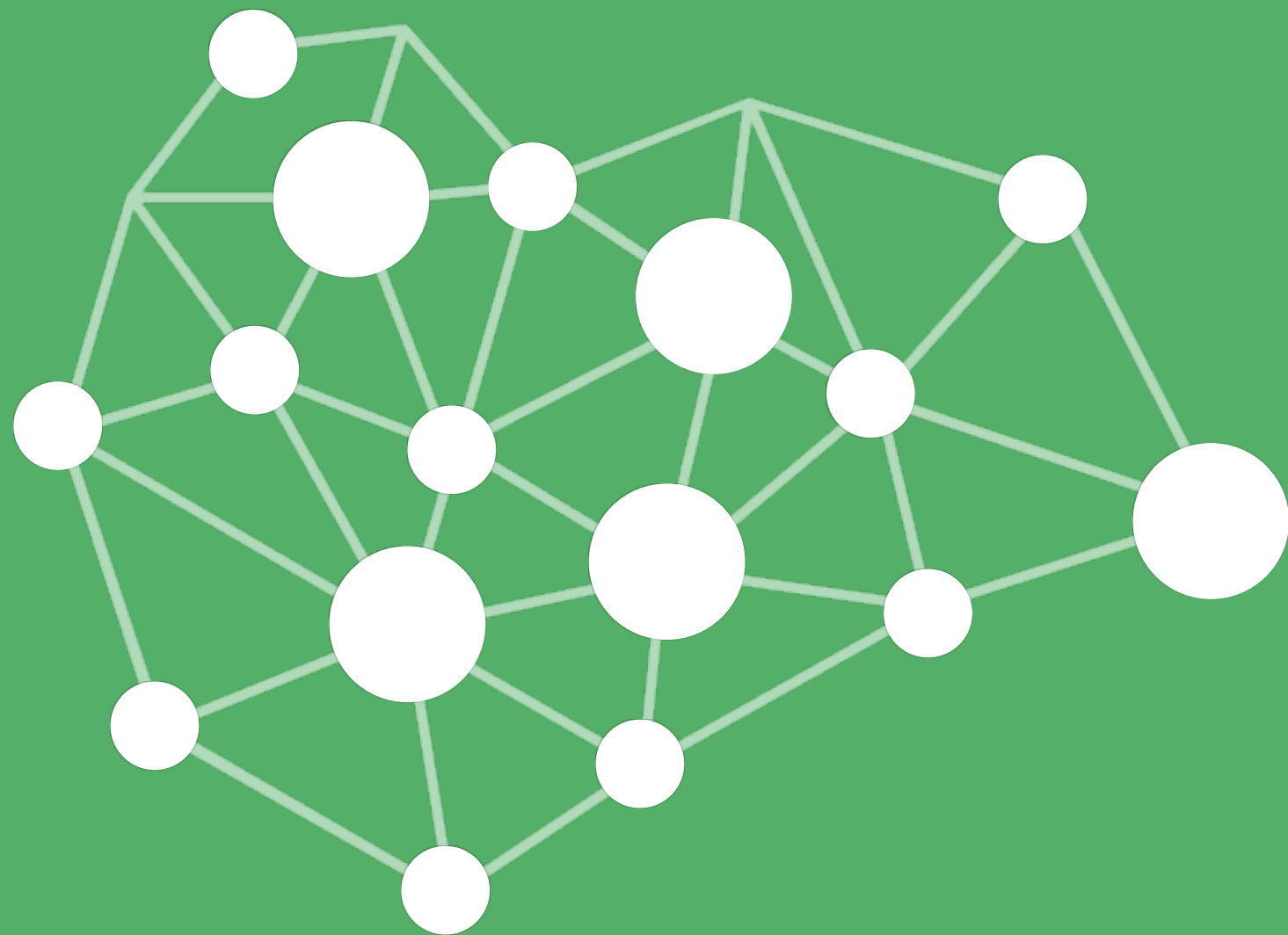
Model Limitations

Models are powerful tools in that they enable us to conceptualize large systems that are beyond our immediate faculties, but they do this by removing certain details. If the model is not built properly it **can blind us to important information required to be successful.**

Models constrain our perception, **they contextualize, frame and condition what we see and don't see.** Models that are poorly built and do not include all relevant perspectives and information can blind us to what would otherwise be obvious. In a very literal manner, models remove one from one's senses in that they are conceptually based, if they are not adequately built they can create nonsensical results.

When we use incomplete models to operate in the world, the results may be, at best, only partially successful and, at worse, a potentially dangerous situation. Thus, it is desirable that we be explicit about the limitations of a model and work to develop more robust models whose foundations are clearer, scope broader and better match the data.





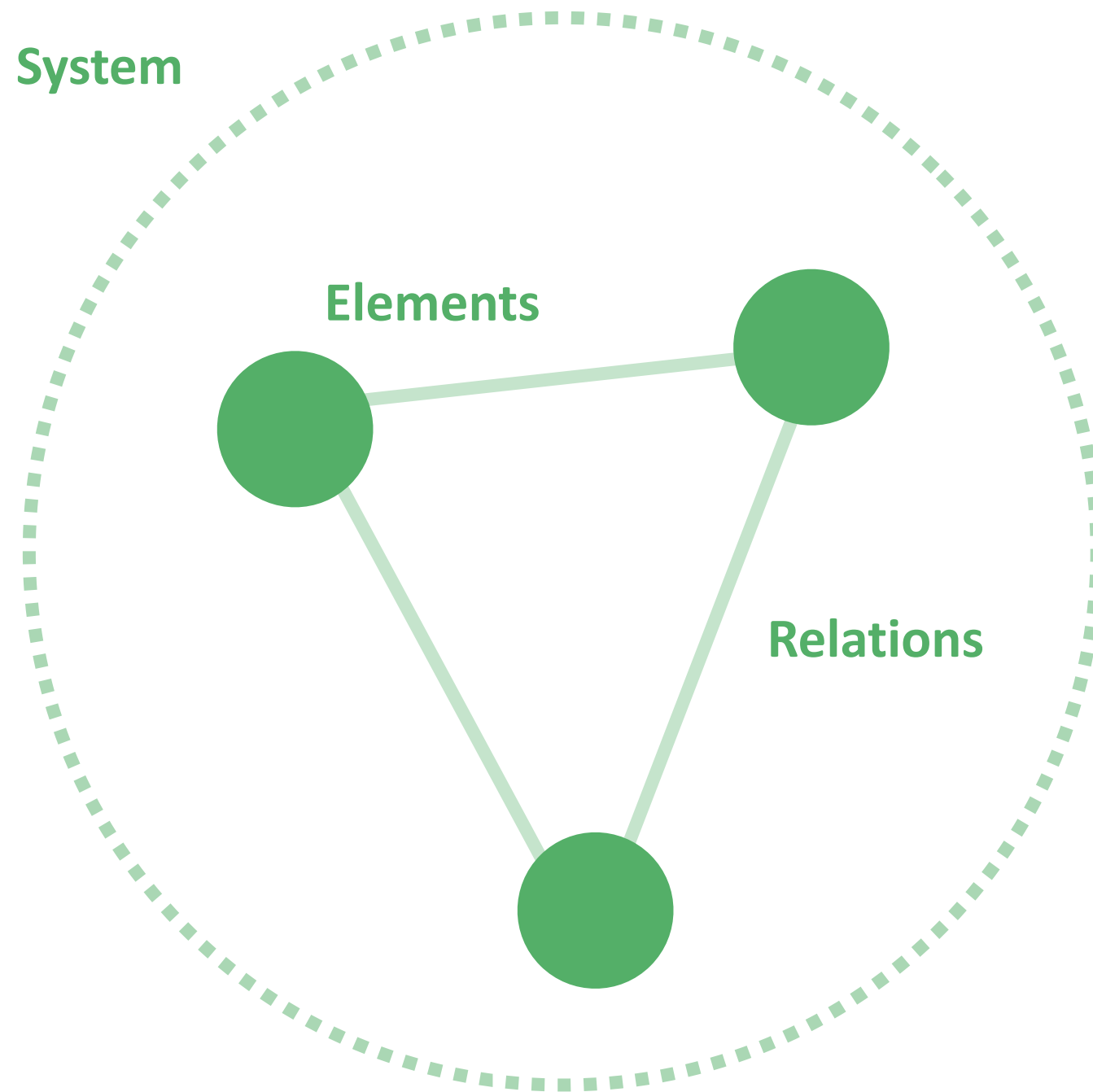
Systems Model

Systems

In this section, we lay down the very basics of our language for describing systems by talking about elements, relations and whole systems. A system may be defined as a set of interacting or interdependent elements forming an integrated whole.

Firstly a system is a group of parts, e.g. it is a composite entity, being composed of a number of things. In the language of systems theory, we call these parts, elements.

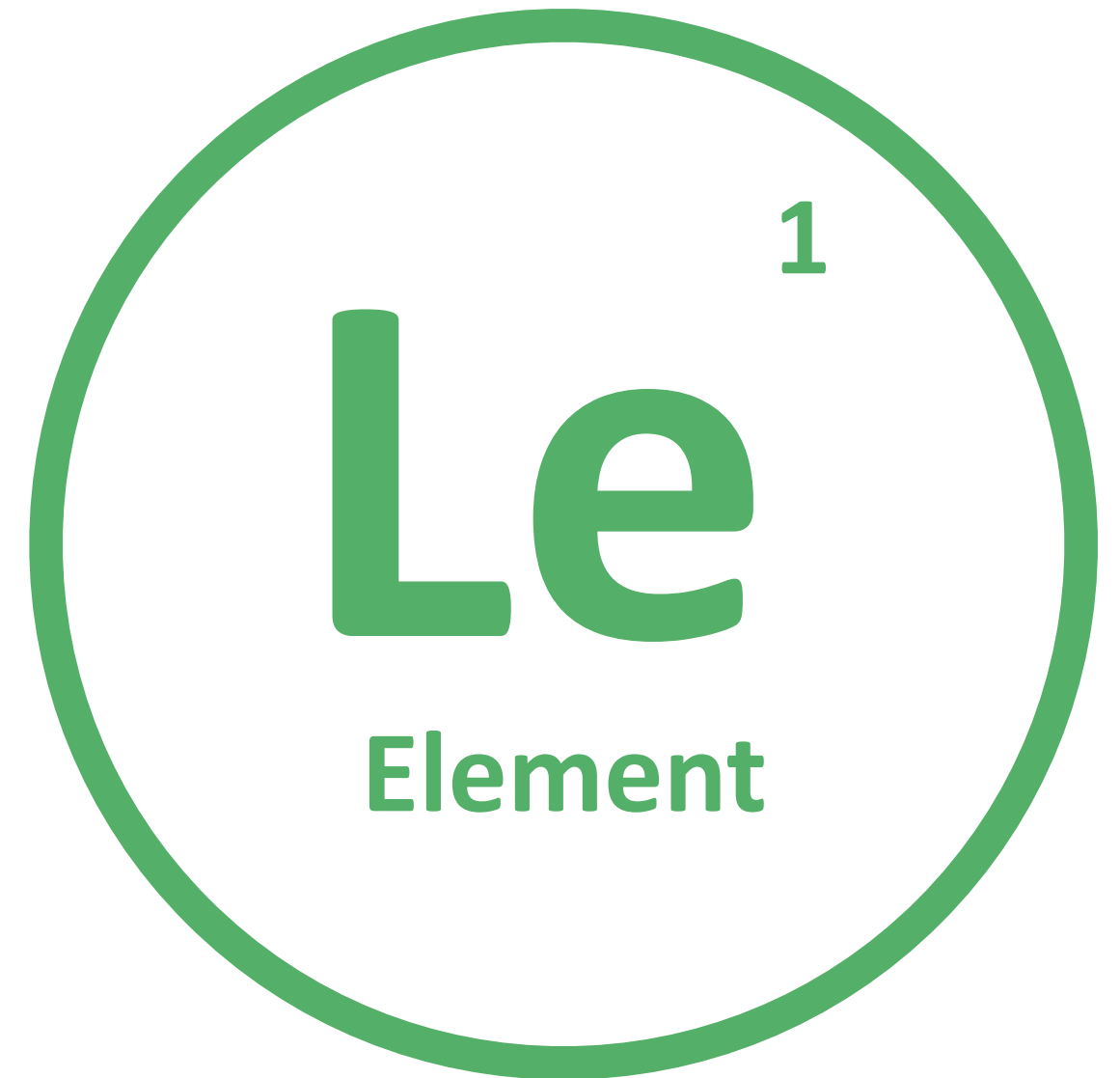
These parts are interconnected and interdependent in some way, that is to say, there is a set of relationships between the elements. Through these relations the elements are arranged in a particular way in order to perform some collective function that defines the system as a whole.



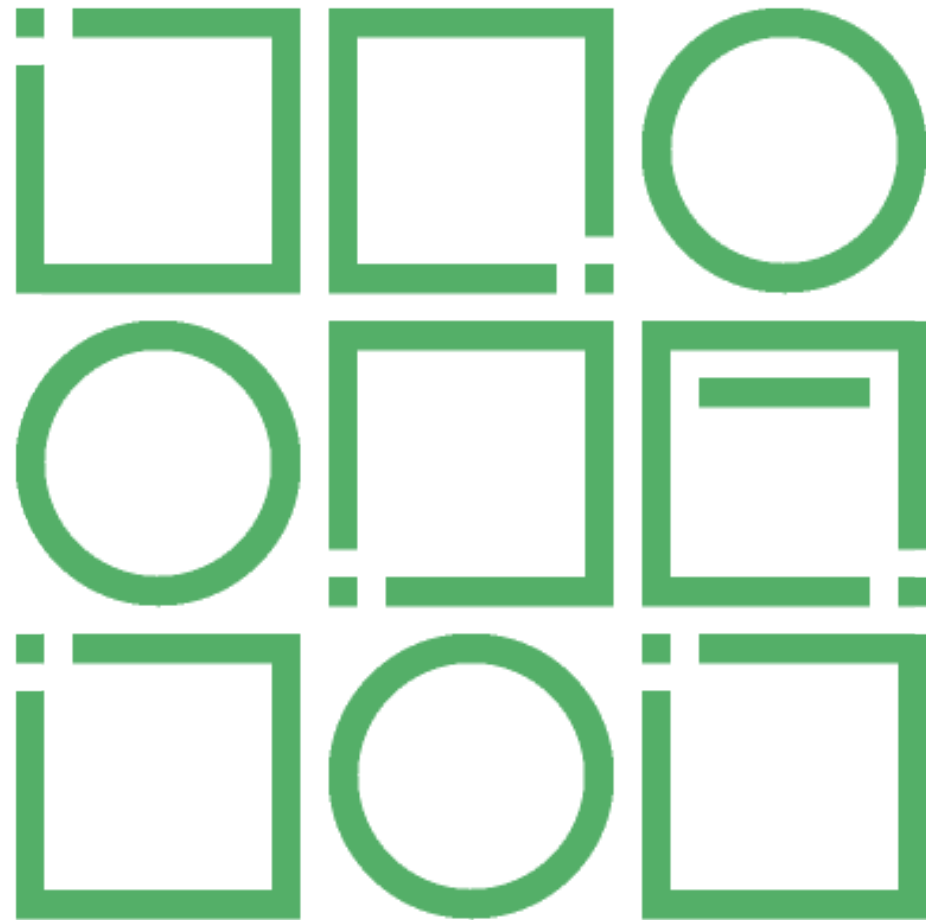
Elements

An element is the most basic unit within the model of a system. Elements are elemental, which means that, with respect to the system of interest, they are not constituted of subcomponents. We can say that elements are black boxes, meaning we do not look inside of an element to see how it works or what it is made of.

An element simply consists of some structure and possibly a function. A stone is an element that has structure but no apparent function. With respect to the study of a whole ecosystem, we may see a plant as an element within that system, as we do not need to consider its internal workings. A bus may also be an example of an element with respect to a transport system. The bus consists of a static structure and a set of mechanical processes that enable it to operate.



Sets



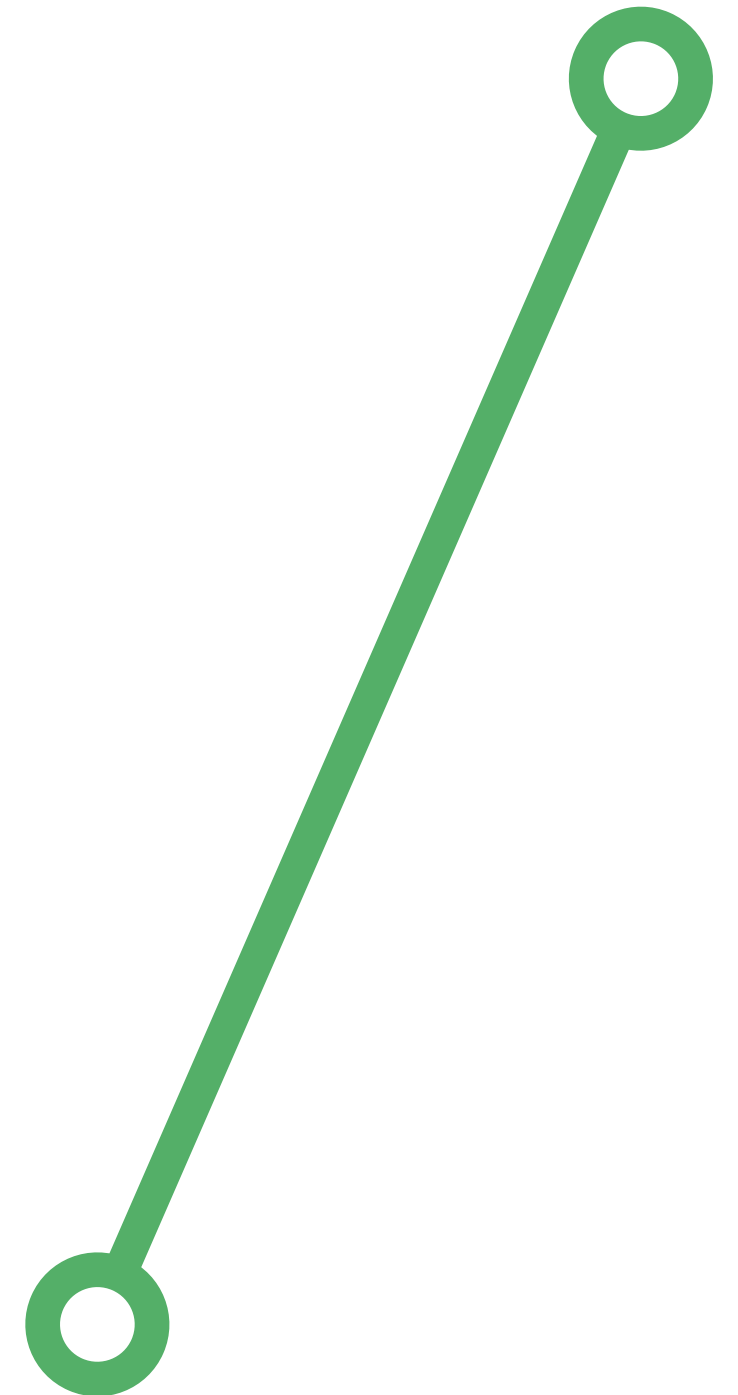
A system is a set of elements that are interrelated to function together, thus not everything is a system. If we have a **group of elements that are not interconnected and do not work together** then this is not a system, it is what we call a set; a simple set of elements.

An example of a set might be a pile of bricks or a group of unassociated people. These parts are independent and have not been designed to work together thus **we describe them by simply describing the properties of each element in the set** and that tells us everything we need to know. There is nothing more to the set than the simple sum of its elements.

Relations

The elements within a system are interconnected and interdependent in some way, e.g., there is a set of relationships between the elements. A relation between two or more elements is defined as an interaction or **connection wherein the variables associated with one become correlated with those of others within the relation.**

For example, the light, pen and cup on a desk do not form a system as they are independent elements; when we change the cup or pen this will not change the other elements. If there is no correlation between two variables then they are not interrelated. In contrast, a family forms a system because there are a set of interrelations between the members; **if the state of one person changes significantly this will change the state of others**, thus they are interdependent.



Type of Relations

These variable correlations may be positive or negative, meaning they can move in the same direction or in opposite directions depending upon the type of connection. A negative correlation is a relationship between two variables in which one variable increases as the other decreases, and vice versa. For example, with two people trading bananas their benefits are negatively correlated, if the seller charges more in the exchange then they will benefit while the buyer will have to pay more. In contrast a positive correlation is a relationship between two variables in which both variables move in the same direction.

Negative Correlation



Positive Correlation



Synergies

Synergies describe how two or more elements interact in a system in either a constructive or destructive fashion. If the elements interact in such a fashion that the **combined effect is greater than the sum of their parts** this is termed a **positive synergy**. If the elements interact in such a fashion that the **combined effect is less than the sum of their parts** this is termed a **negative synergy**.

Negative Synergy



Positive Synergy

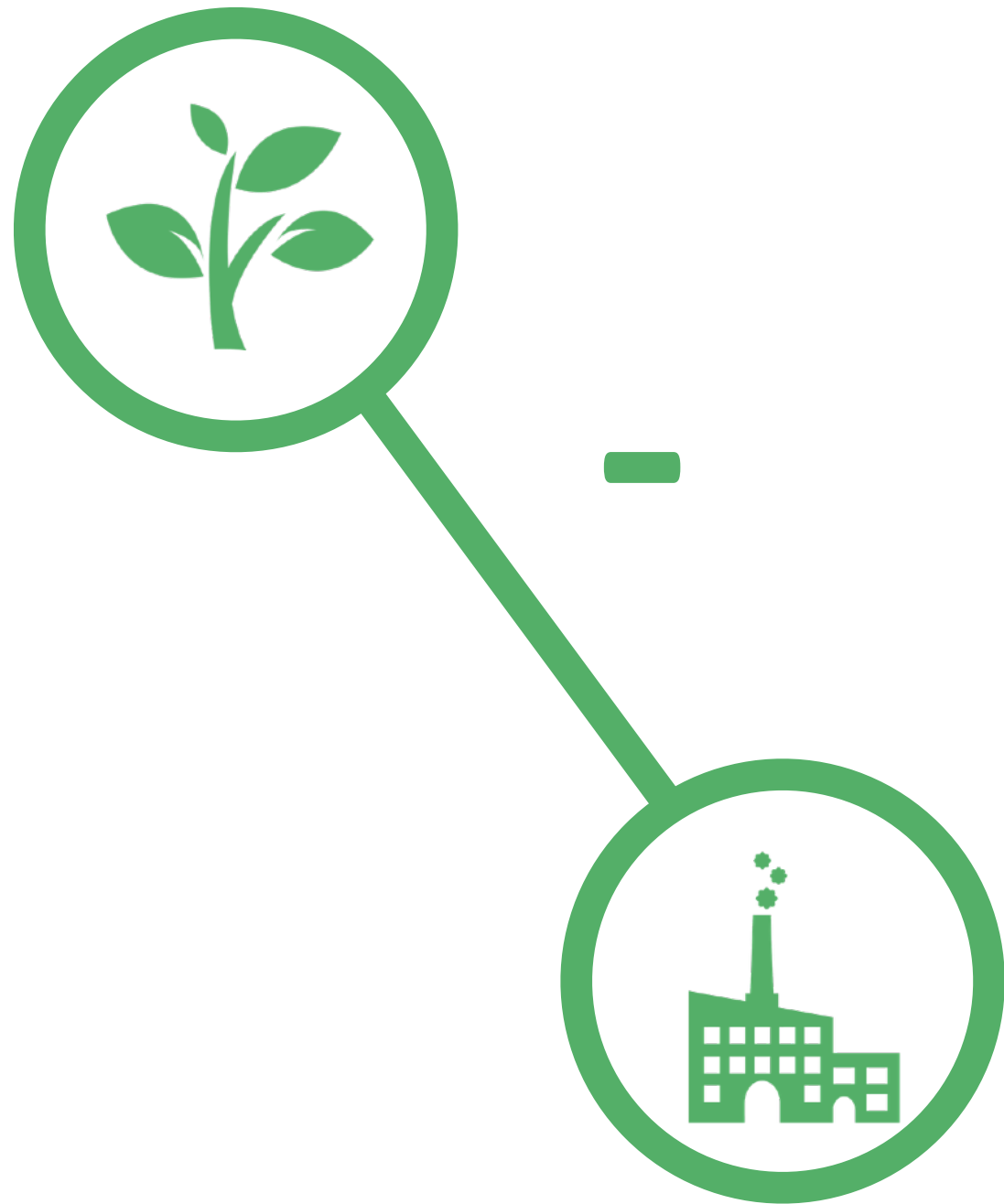


Negative Synergies

A negative synergy is a destructive interaction between elements; **a relation of interference**. A negative synergy is when two elements interact and the combined effect is less than the sum of their effects in isolation.

The relation between two drugs can be a negative one, where when prescribed together they interfere with each other resulting in an overall counter-productive effect.

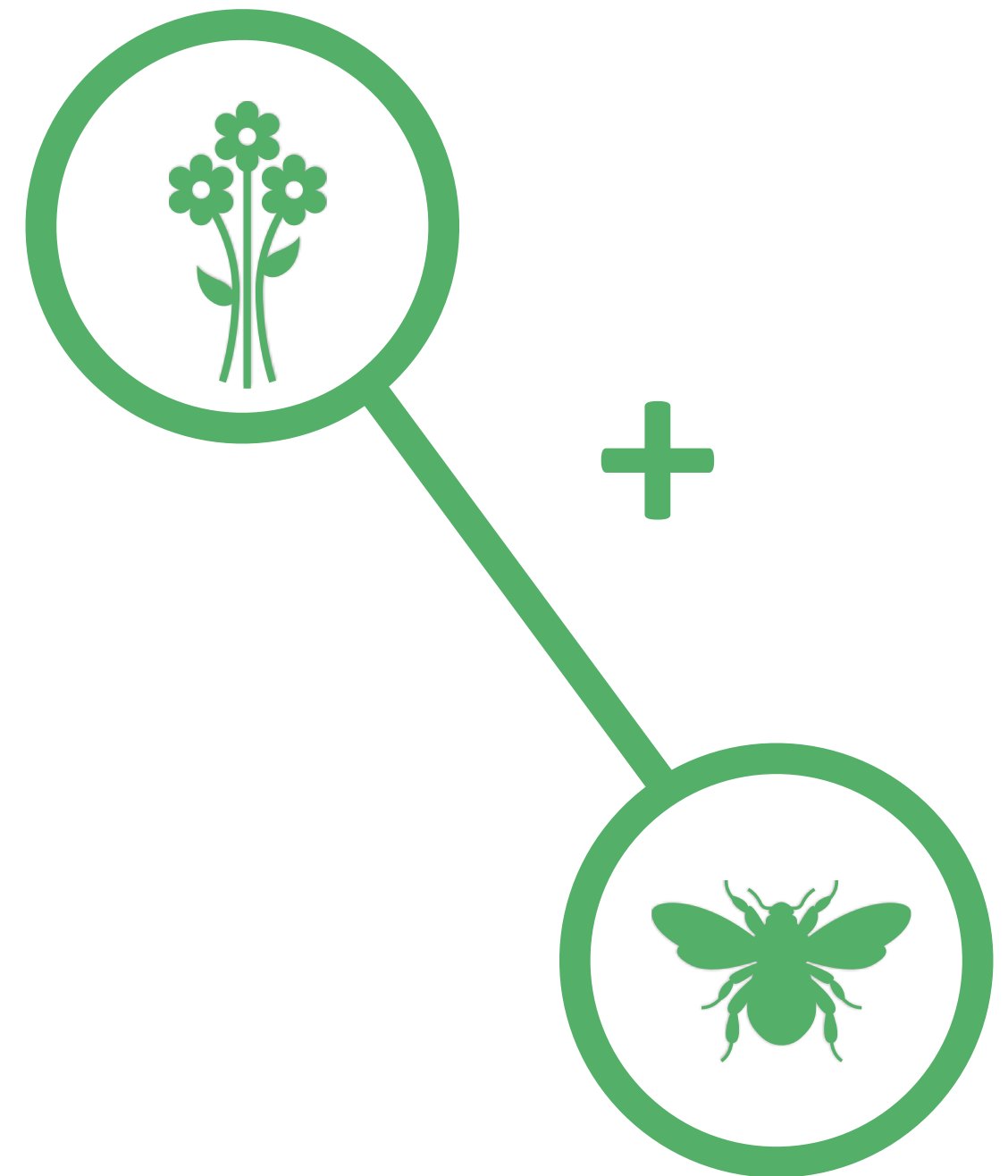
Another example might be two businesses within a given market sector, they may engage in a competitive relationship thus creating a dynamic of interference between them. In contrast, they may choose to cooperate and through this synergistic relation enhance the overall state of the market, this would be a positive synergy.



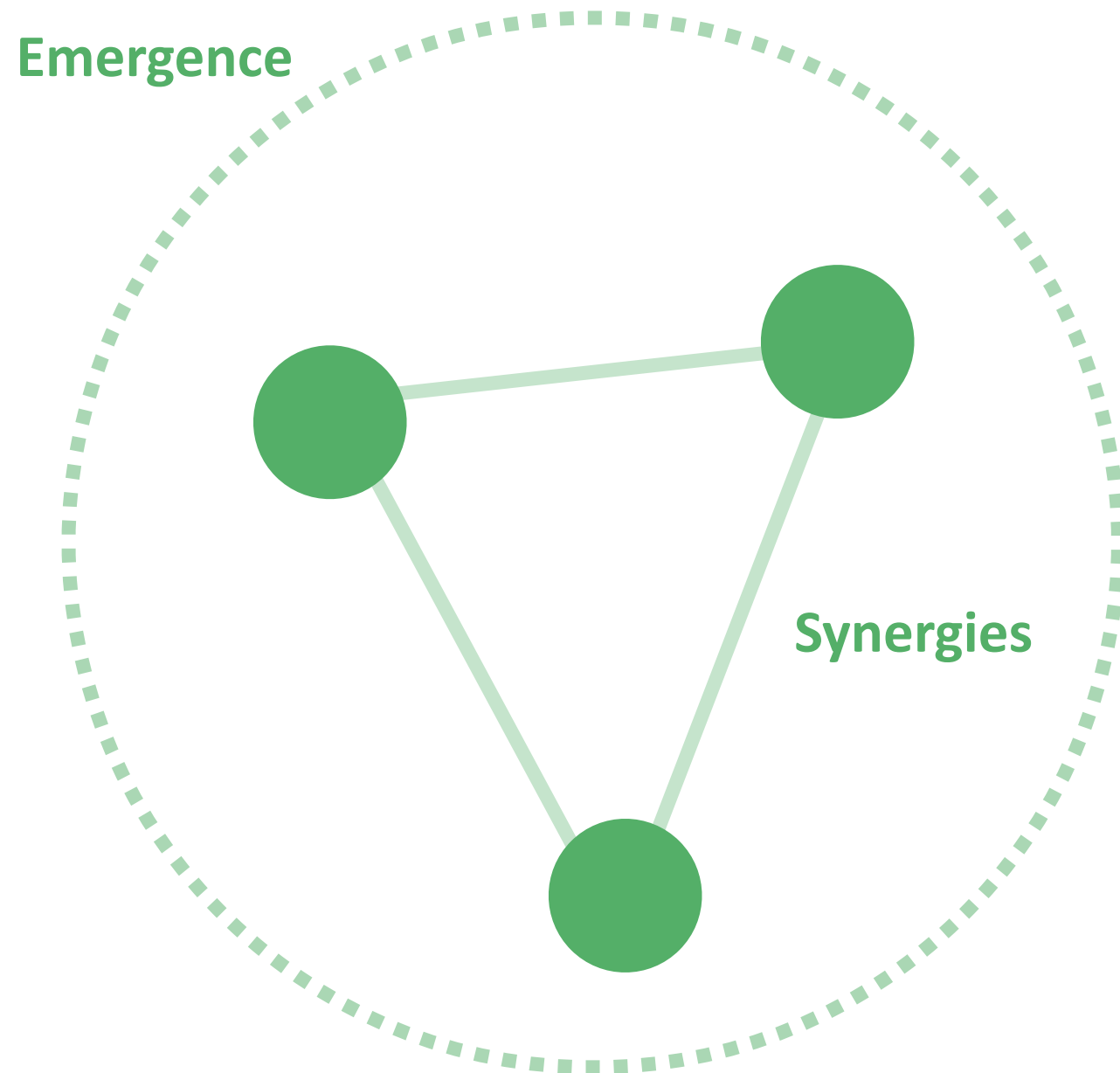
Positive Synergy

The term synergy comes from the Greek word "synergia" meaning "working together". A positive synergy is the exchange of energy or resources between elements in a constructor fashion where **they work together to perform a common function**. By working together through synergistic relations the entire system may become more than the simple sum of its parts in isolation.

The division of labor within many animal and human communities could be cited as another example; ant colonies and advanced market economies are examples of synergistic relations where **the net result is greater than the product of the individual elements acting in isolation**.



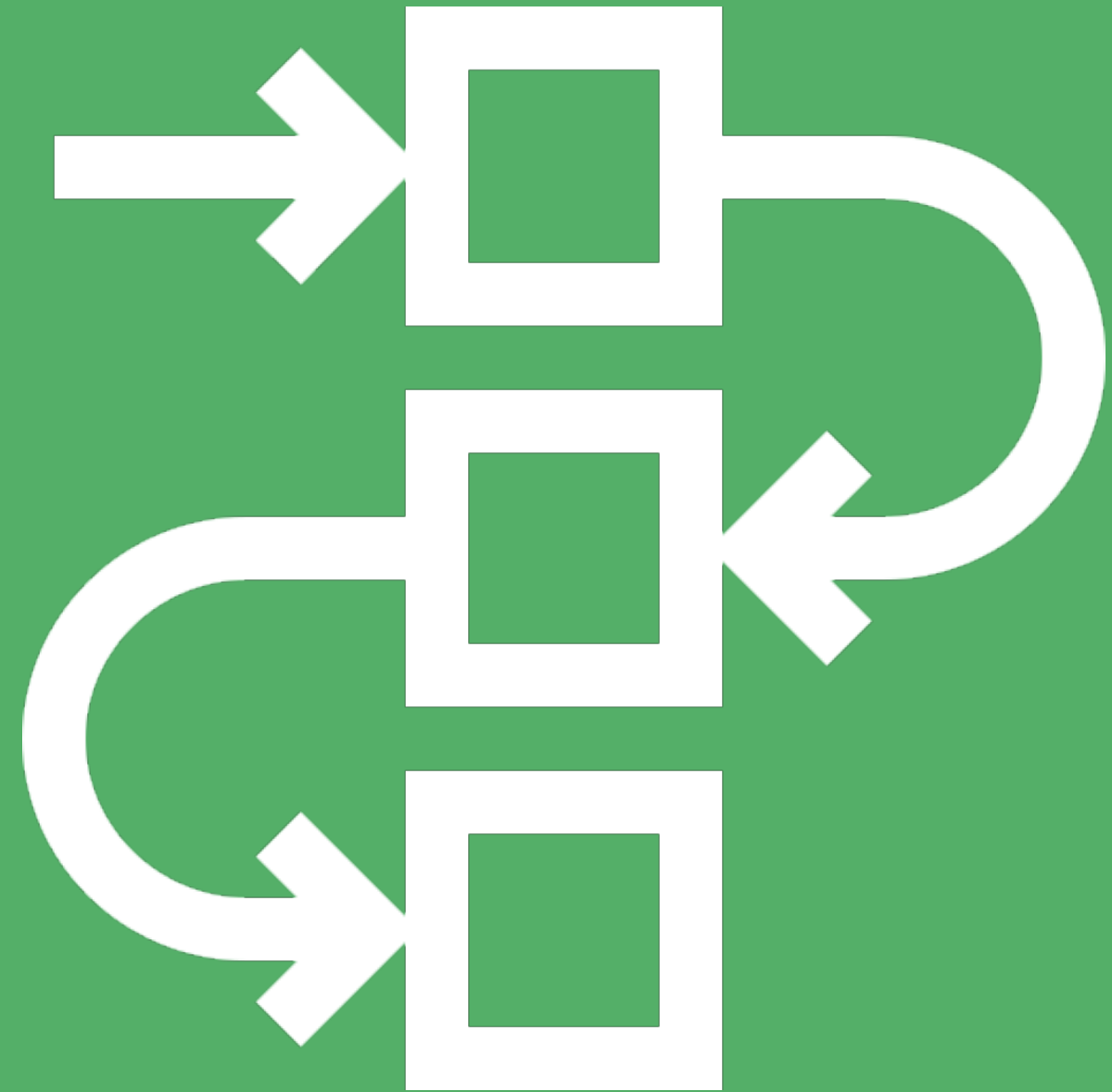
Emergent System



A system is what emerges out of the interaction between the elements when they work together as an entirety. There forms a whole system with a macro-scale structure that is an emergent property of the organization of elements and through these interrelations the elements can come to share some common function.

For example, the human body is a system because its organs - the system's elements - are interconnected and interdependent in serving the overall function of the body. In order for the whole body to function as an entirety each organ has to be specifically designed in relation to the other parts to achieve a synergy between them and the emergence of the whole as a functional system.

Systems Functioning



Overview

"To enable a system to perform effectively we must understand it - we must be able to explain its behavior - and this requires being aware of its function in the larger system of which it is a part"

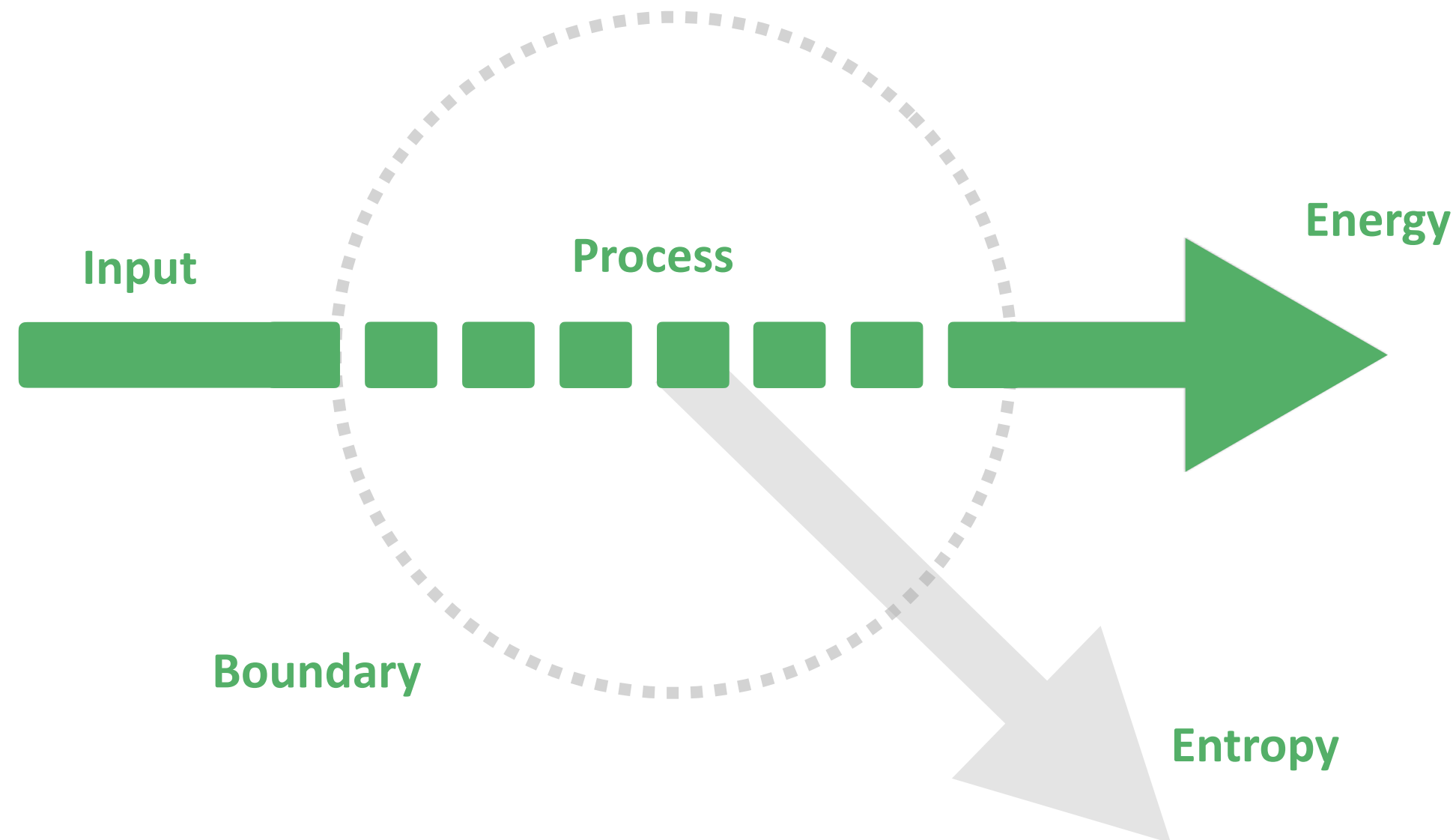
- Russell L. Ackoff

A central aspect of systems modeling is to define the systems function. **Systems perform functions as an entirety and this defines much of how they are organized and operate.** In this section, we are going to build upon the model we have so far created to introduce the process through which the system takes in inputs and maps these to outputs through a set of stages.

Through a process, **systems take in resources and perform a set of operations on them to generate an output.** The output of the system can be of benefit to other systems and its environment - what we may call energy - or may be detrimental - what we call entropy. Understanding and defining the **systems processing is important in telling us what is a functional system vs a dysfunctional system;** what is of value and what is not of value.

Function

A function is a **very broad and fundamental concept** that is central to systems theory, it is also used in many other domains being particularly important within mathematics and engineering. **A function is a process that transforms energy or resources from one state to another.** There are a number of key aspects to this process. We have a resource that is inputted, we have a process that transforms these resources according to a set of instructions to create an output.





Input

Inputs involve the capturing and assembling of elements that enter the system to be processed. The water taken in by the roots of a plant is an example of an input to a biological system, the fuel inputted to an airplane is an example of an input to a mechanical system.

Any given system can only process a specific range of inputs. What a system can and cannot process is defined by its function. Biological systems process natural resources, mechanical systems process fuels, information systems process data, etc.

In this respect systems exist in relation to their function and what they can process; a plant exists in relation to the natural environment because that is where it gets its resources from; a hard drive exists in relation to a computer, because that is what it exchanges resources with.

Processing

Resources that are inputted are processed within the system. A **process is a series of actions performed upon the input** in order to achieve a particular end result, e.g., the output. A process alters or changes the resources according to some set of instructions. Systems contain some logic or information as to how to transform their inputs into outputs.

The **workings of a process is defined by an algorithm** or set of instructions that are performed upon the input in order to produce the output. The function of running a factory or the workings of a political system follow a certain logic that is somewhat algorithmic in nature. In the case of the political system this is encoded in the constitution, or for the factory in a production process design. From this we can see how systems have a certain logic that is defined by their operations. **Understanding this logic to the systems operation is key to understanding why and how they behave.**

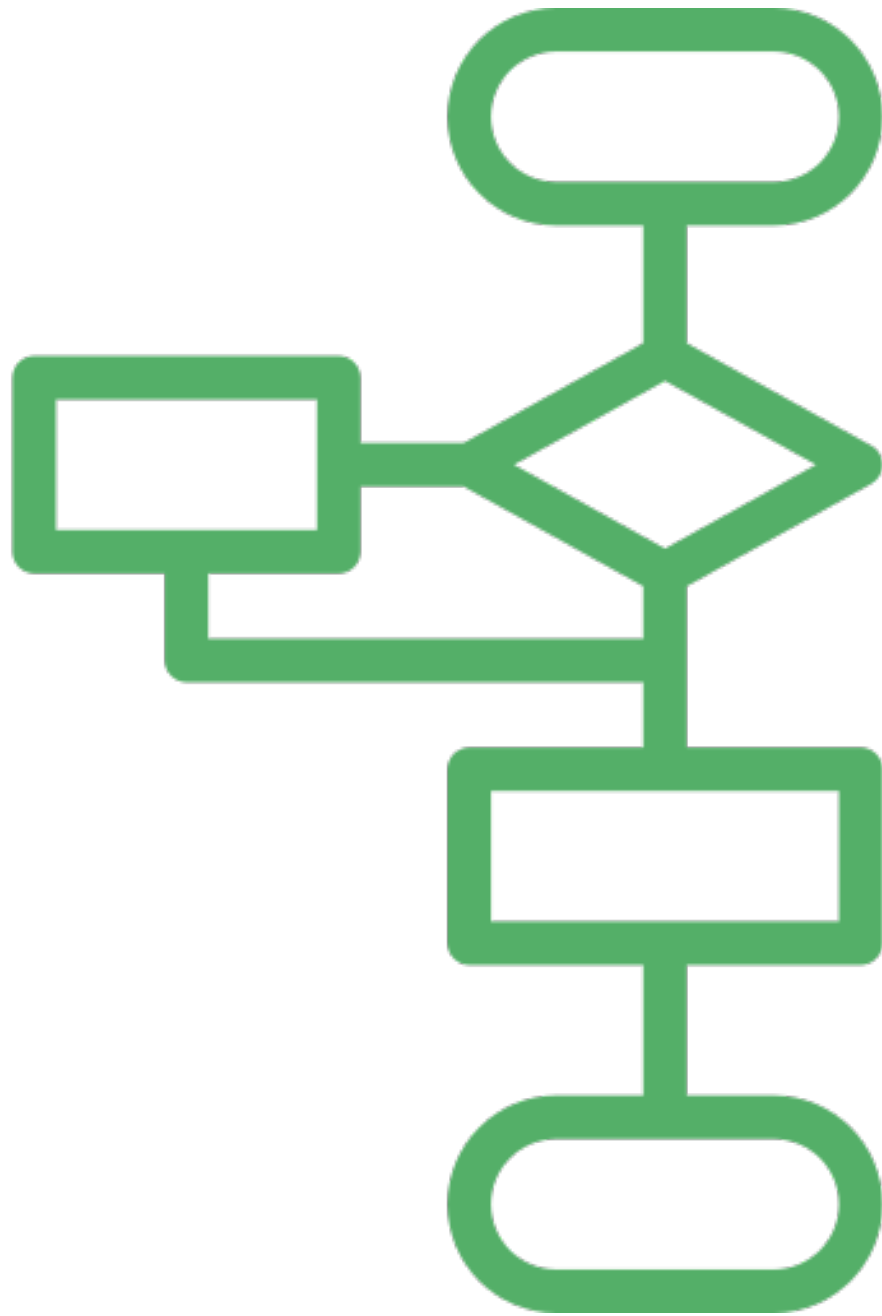


Types of processes

The internal workings of a process can take many forms, from simple to complex - from the simple washing of dishes to the complex process through which a plant performs photosynthesis. Although often represented in a linear fashion, for simplification, **processes are often nonlinear being cyclical**; feeding back on each other with the output from one process being the input for another. Processes may be also nested within larger processes or run parallel to them.

Functions may be unidirectional, meaning the function only maps an input to an output, or bidirectional meaning it can also invert this process to transform the output back to the input through what is called an inverse function.

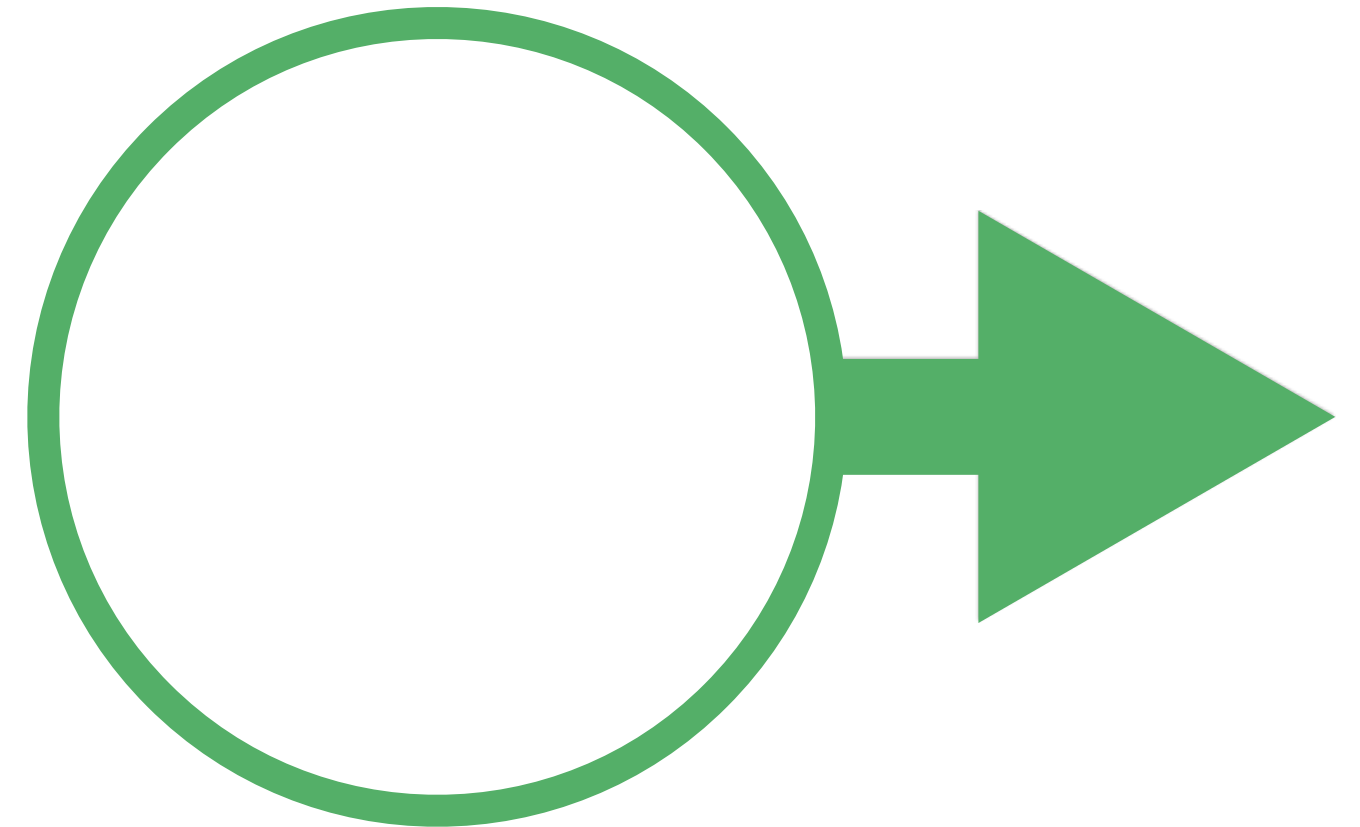
Many processes are essentially unidirectional requiring vastly or even infinitely more energy to invert the function than was required to perform it in the first place. The building of a Lego brick house is an example of a function that can be easily inverted and thus bidirectional.



Output

Ultimately there will be some energy or resources produced by this process that travels across the system boundary to be returned to its environment and this is what is called the system's output. One way of understanding a function then is simply as the difference between what goes into the system and what comes out.

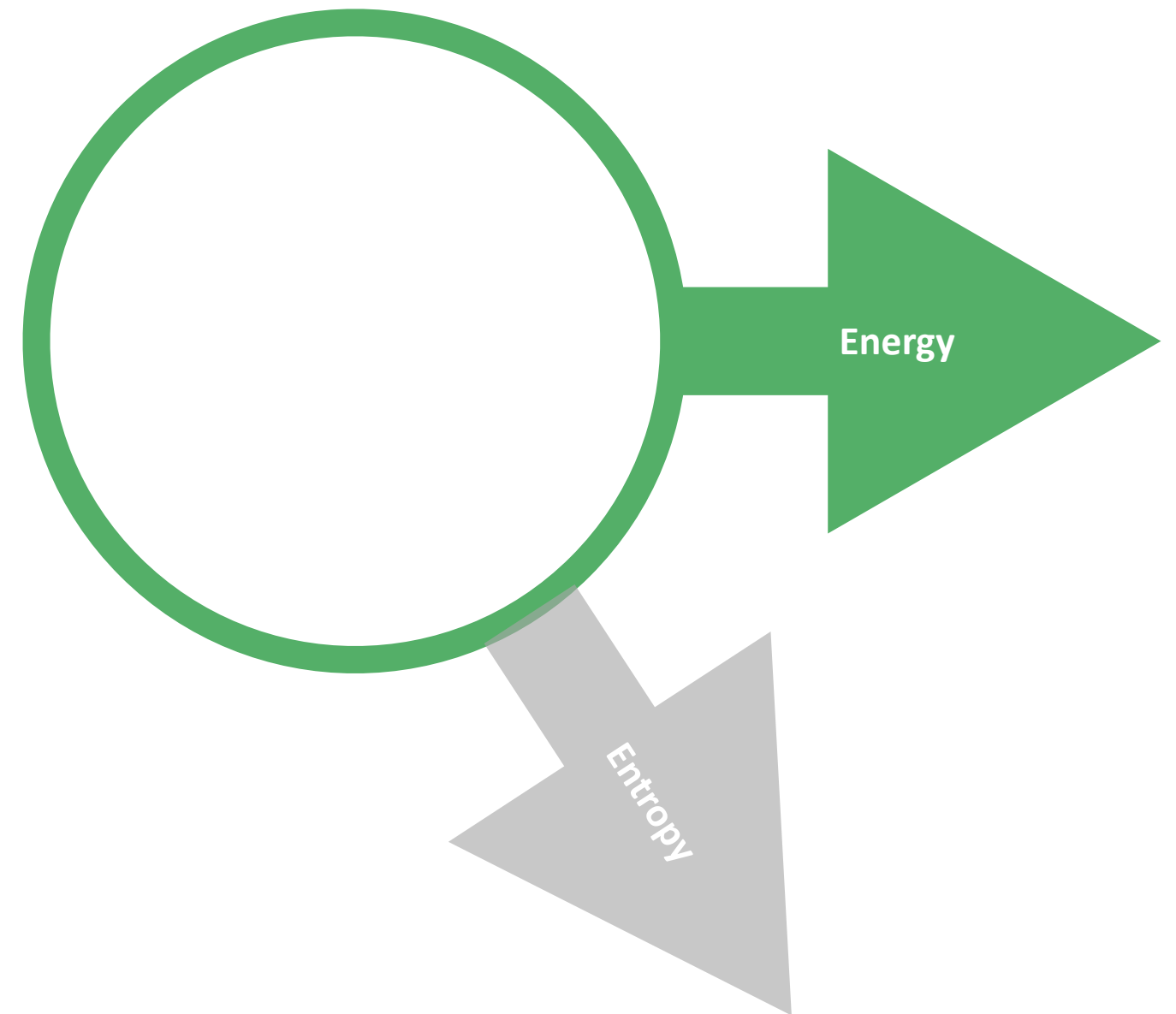
If we can see the internal workings of the system's operation we can call it a "white box" system. While a "black box" system is one that is viewed in terms of its inputs and outputs, without any knowledge of its internal workings. Black boxing can be of great value to us as it helps to hide away the complexity of the internal workings of the system.



Energy & Entropy

The output to a system can take two qualitatively different forms depending on if it is of value to its environment or detrimental to its environment. An output that is of value to other systems in enabling them to do work we may term energy or a resource. Energy can be understood as anything that provides a system with the capacity to do work or to build up a functional structure.

The opposite of energy is entropy, which is the incapacity to perform work and a measurement of the degree of disorder within a system. A dysfunctional output that is detrimental to other systems may be termed entropy. Entropy may be loosely equated to the term waste. For example, a haymaking machine may perform a function of producing bales of hay, but it may also produce noise pollution which we would term entropy. Entropy reduces the capacity of other systems to function.



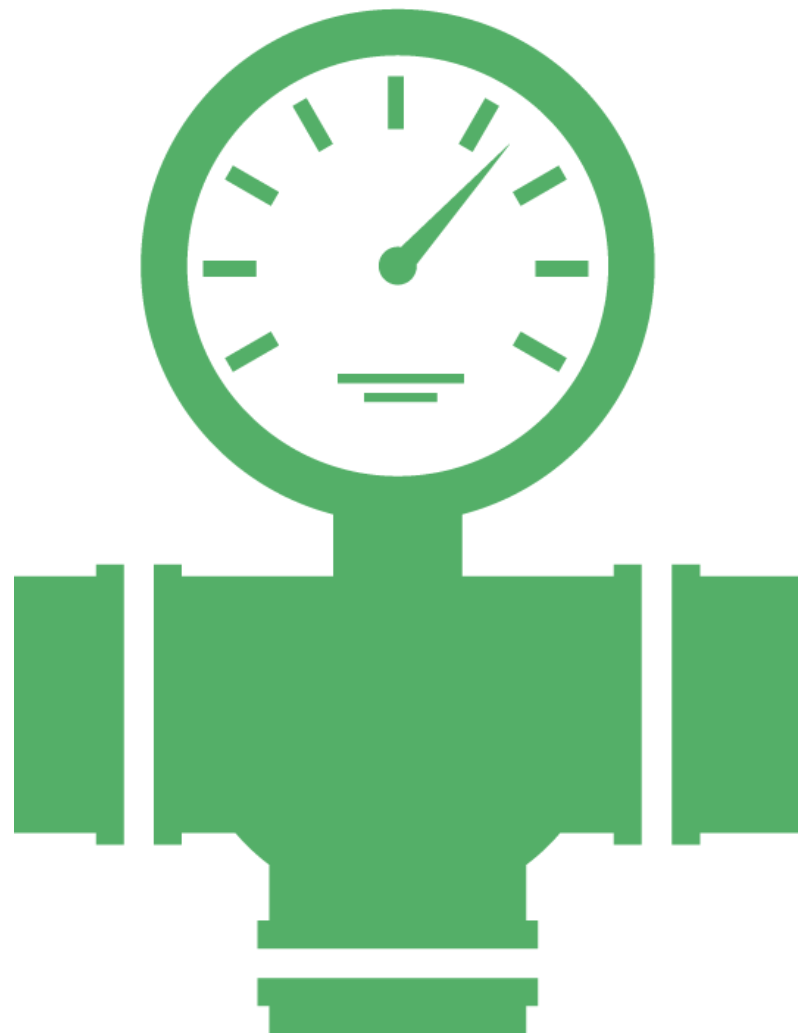
Efficiency

Efficiency is the ratio between the resources that a system processes into a functional output of some energy resource versus the output of entropy or waste. For example, many of our engineered systems are measured in terms of this type of efficiency, such as the energy efficiency of a light bulb or fuel efficiency of a car.

Outside of engineering another example might be an economy that takes in some amount of resources - natural, human, financial, etc. - and generates both a functional output in terms of developing the economy - better infrastructure, more human capital, greater financial capital, etc. - but also dissipates some of the resources inputted. The efficiency of the overall system would then be the ratio between the two that can be expressed as a rational number.



Consumption vs Production



The efficiency of the system's functioning defines whether the system is efficient and productive or inefficient and dysfunctional. A system operates through the input of some resources from its environment with this input then being processed in one of two ways; either productively processed or dissipated.

The system may perform a constructive function on the input thus outputting a resource of a greater or higher potential energy back to its environment. A function can be said to be productive if the system takes in some resource from its environment and performs work on this resource by transferring energy to it and thus outputting a resource of greater value.

Inversely the system may perform a dissipative operation; conserving the energy resource within the system's boundary and outputting the resource at a lower state of potential energy. A system can be said to be inefficient when the resource that was inputted transfers its energy to the system; conserving this energy within the system's boundary whilst outputting entropy to its environment, e.g. political corruption.

Functional & Inert Regimes

Systems that are low in efficiency we may term "inert" which is the incapacity to perform work. In this regime, a system dissipates more energy than it converts to a functional output and is thus defined by what it consumes. For example, Industrial economies have come to be termed "consumer societies" because they input large amounts of resources while producing high levels of waste through dissipation.

As another example, a person may use the resources they consume during a day to go to work and perform some function within their society or they may stay at home playing computer games. With respect to the individuals environment - the economy - they have dissipated the resources they consumed while returning waste back to their environment.

An efficient system occupies a functional regime defined more by the function that it performs. For example, a person that works hard will be define by the work that they do rather than what they consume.





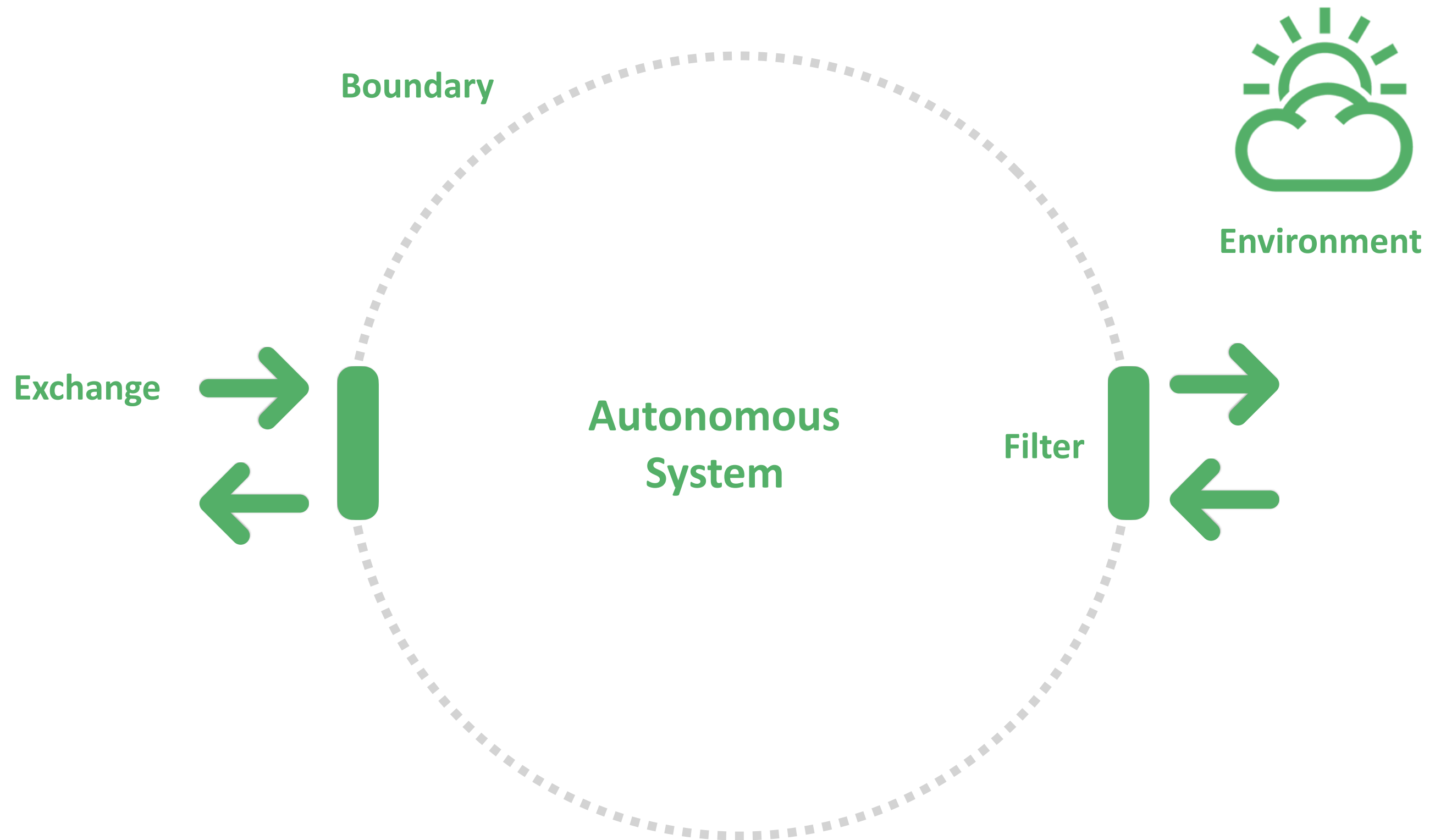
System Environment

Overview

All systems have a boundary and operate within an environment. This environment represents the sum total of other systems and input/output resources that the system interacts with during its operation. Thus the environment consists of the sum total of resources and systems that lie outside of the boundary of the system of interest and interact with it providing its inputs and receive its outputs.

A system's environment is primarily relative to its functioning. A biological system that requires the input and output of natural resources operates within the natural environment. The political system of a nation operates within the context of that society and the international political environment.

An environment is the broadest unit of analysis representing the sum total of all systems and interactions that we are considering within any given model. As the environment represents the sum total of the system's interactions with other entities, it forms the context to a system, i.e. where it is in relation to other systems that it interacts with.



System Boundary

Systems have limits, and these limits to a system we may define as their boundary. The system boundary demarcates a limit to the system's internal components, relations, and processes. Internal to its boundary the system has some degree of integrity, meaning the parts are working together and this integrity gives the system a degree of autonomy.

If we take a small plant, for example, the parts to the plant are all interrelated in performing its overall operations and through this integration, the plant is able to exist autonomously - to some degree - from other systems in its environment.

Boundaries may have a physical dimension but can't always be defined in physical terms, for example, we may be talking about an information system or a social system. If we want to be able to achieve sufficient generality to talk about all types of systems - which is the aim of systems theory - then we need to understand boundaries within this slightly more abstract language of integrity and autonomy.

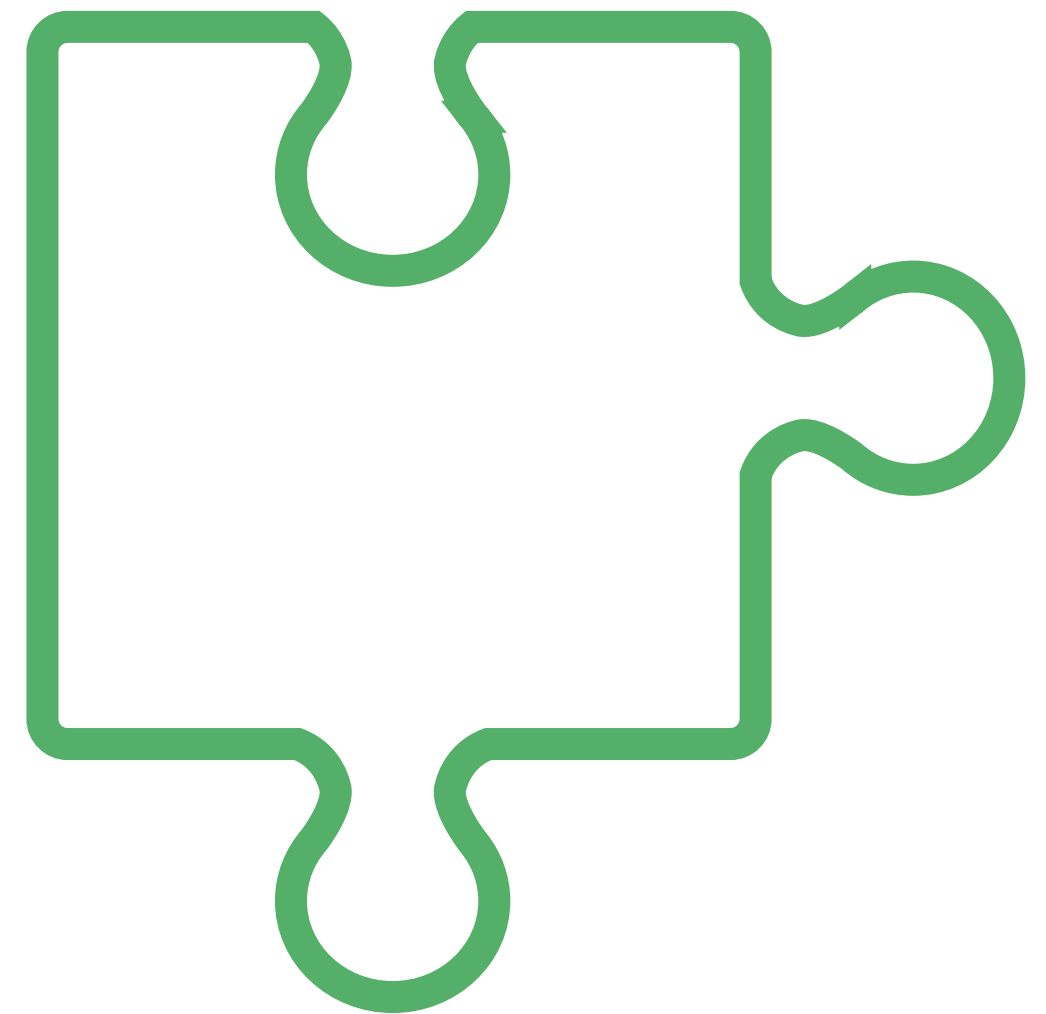


Autonomy

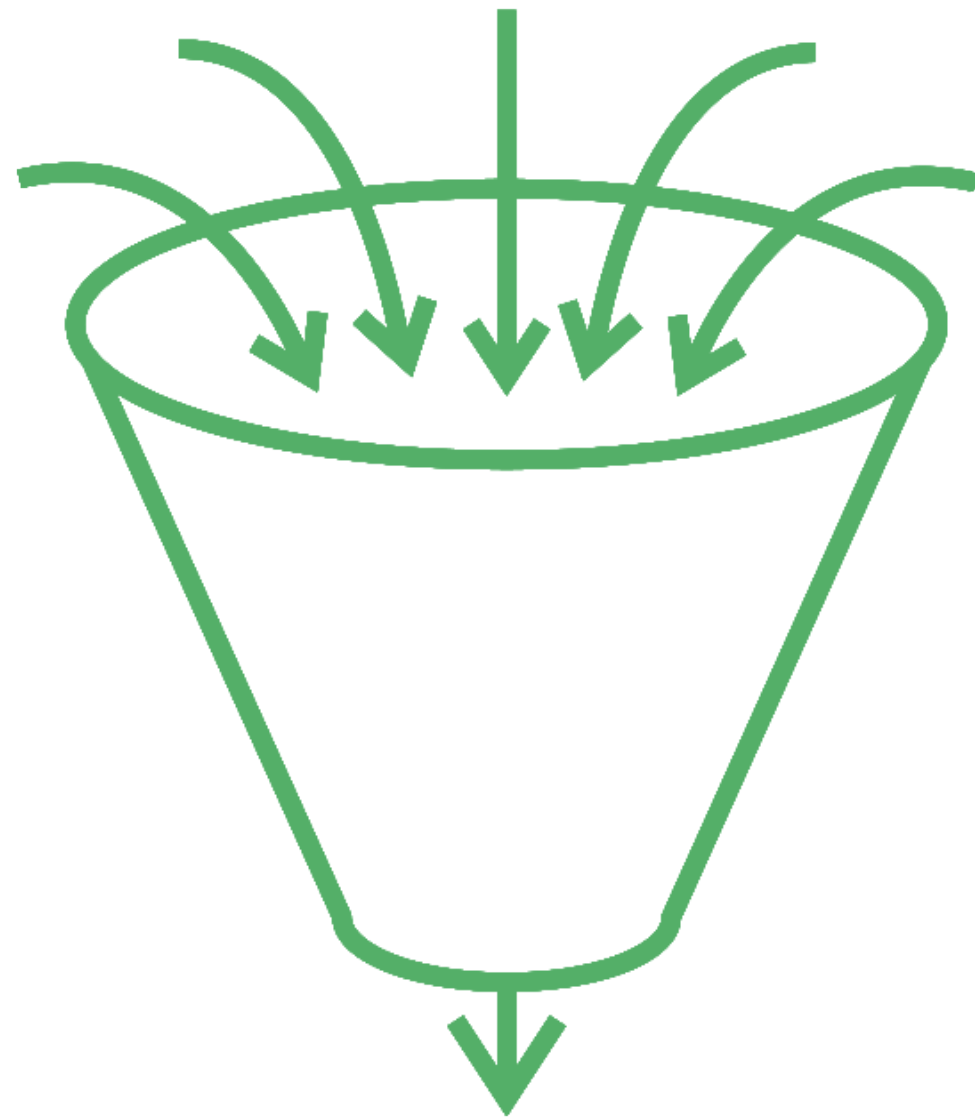
A system's boundary is then demarcated by where the **nexus of relations that enable it to function as an integrated and autonomous whole reach their limit**. Beyond this, the system loses its autonomy and has to interact with other systems and its environment.

A system is always dependent upon its environment in some way. The more inefficient and consumptive the system is the more dependent it is upon its environment to provide it with the resources to operate, thus there is a direct **connection between systems functionality, dependence, and autonomy**.

The boundary to a nation-state is another example of this, the nation's border is only a boundary if within this boundary public functions are integrated within the national system as an entirety. By the nation functioning as a whole, it can be autonomous from other nations.



Filtering



Any given system can only process a specific range of inputs. **The system boundary functions to filter inputs to the system.** These filters often function to prevent damaging or harmful inputs while being receptive to those that are beneficial to its functionality. For example, our skin functions as the boundary to our body and the human immune system as a filter to prevent harmful antigens from entering.

Systems interact with other systems through protocols. These are specific **channels, mediums and rules through which the system can successfully interoperate with other systems.** For example, as an English speaker we can successfully interoperate with other English speakers because we know the protocols of communications of this language, however, many of us cannot successfully communicate with Japanese people because we do not know the protocols required for that interaction.

Open & Closed Systems

Within the language of systems theory, systems are said to be open, closed or isolated. Open systems interface and interact with their environment by receiving inputs and delivering outputs via their boundary. **With open systems the boundaries are permeable**, meaning that they may permit the exchange of materials, energy, information or ideas. Living systems are examples of open systems as they must import and export energy and entropy to and from their environment. Likewise all complex systems are open systems, indeed, most of the systems we are concerned with in our world are open dynamic systems.

Closed systems have limited interaction with their environment; a strong boundary condition that resists exchange. An isolated system is more restrictive than a closed system as it does not interact with its surroundings in any way. The universe as an entirety might be an example of an isolated system, but it is debatable as to whether such a construct could exist in reality.



Acceptance & Resistance

A shop is an example of an open system as it is continuously taking in new customers while others leave; receiving supplies and removing waste, while also communicating with other systems, such as suppliers and their bank. This high rate of input and output to open systems makes them dynamic; they are constantly changing and have to respond to the changes within their environment.

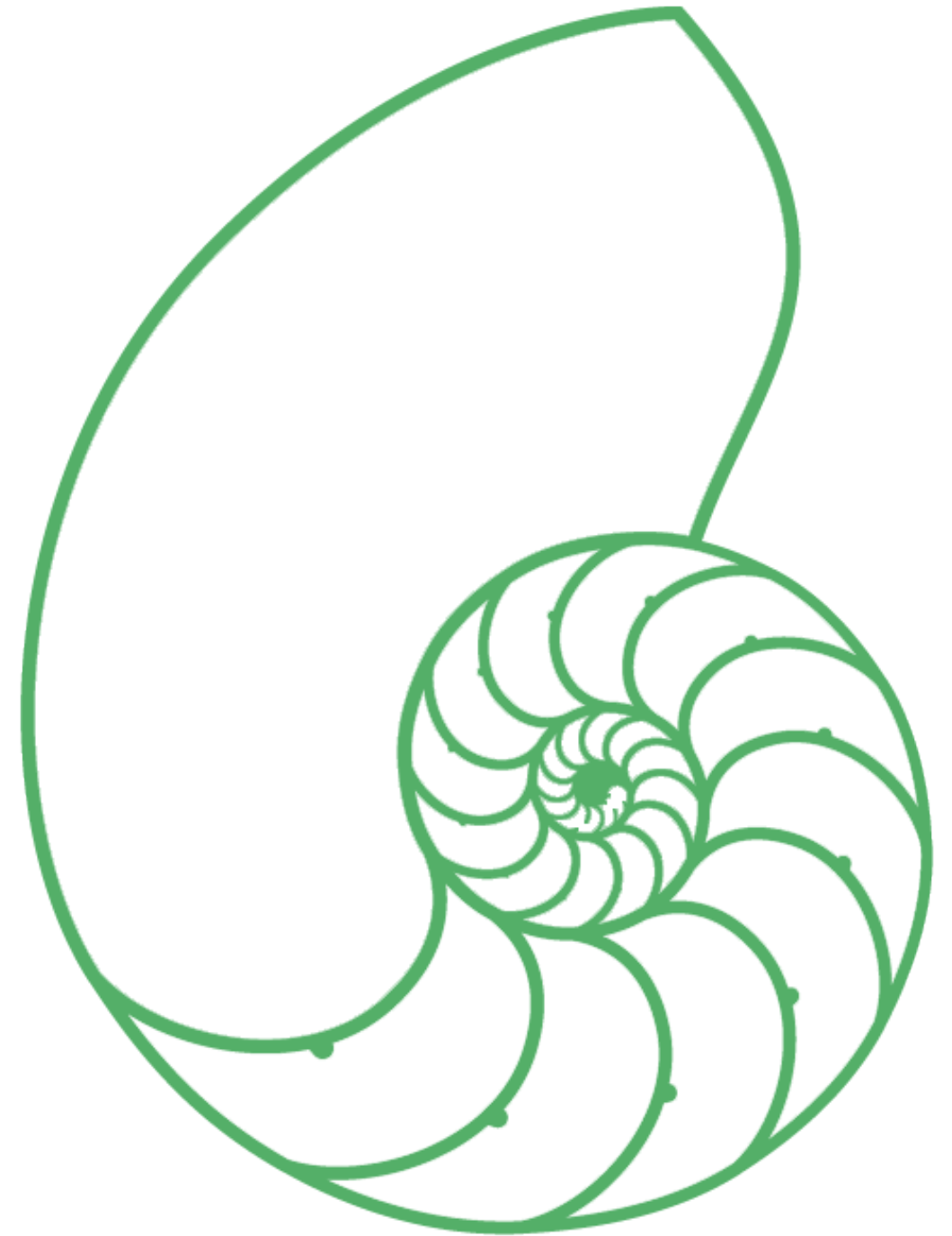
Inversely, closed systems are more prone to resist incorporating new inputs, and this resistance at their boundary makes them more strongly defined by the static properties of their boundary. By not adopting inputs, a closed system ceases to properly serve a function within its environment and may become deemed unnecessary to its parent environment risking atrophy. However there is nothing inherently positive or negative about a system being closed or open, such claims are dependent on the context.



Hierarchical Organization

Systems that are complex exhibit many levels of organization, from the very smallest to the largest levels. The primary way that **systems structure this complexity is through the use of hierarchical abstraction**. Almost all complex systems exhibit hierarchical structure; whether we are talking about physical systems, where atoms makeup molecules which makeup substances and so on, or social structures where individuals makeup organizations which makeup societies, etc.

This hierarchical structure is achieved through encapsulation where **smaller more specific systems are encapsulated within larger more general systems**. For example, the internal operations of each cell within our body are encapsulated within a cell membrane, that abstracts away the details through encapsulation. These cells then are combined together into tissues and organs, **with each new level having new emergent properties and functions**.



Systems of Systems

In order to structure our language when talking about these different levels of the system hierarchy we have five different terms; elements, sub-systems, systems, systems of systems and environments.

At the most basic level of the hierarchy are elements, which are elemental, meaning they do not have constituent components, we treat them as a whole. An electron is an example of an element, we cannot look inside of it because it is not made of any separate parts.

A subsystem is a set of elements, which make up a system. An example of a subsystem might be the breaks in a car, they are made up of elements but are also an integral part of a broader system, the car. The car, which is a system of personal mobility, is in turn part of a transportation system and we call this level to our analysis a system-of-systems. Lastly, all of this is encapsulated within our ultimate unit of analysis, which is the system's environment.



Micro-Macro

With each new hierarchical level to a system, new functions and processes emerge on that level. In order for those functions to be fulfilled properly, the whole organization needs to in some way assert a downward effect on the subsystems to ensure they operate in the way that is required to fulfill the higher-level functions, while at the same time the parts have their own goals thus emerging a complex dynamic between bottom-up and top-down ordering. This dynamic emerges within all complex systems.

For example, in society, we have legal systems that regulate individuals and groups to ensure they comply with rules that enable the whole of society to operate successfully. While at the same time those individuals wish to retain their autonomy and freedom to pursue their own ends. The result is a control or regulatory dynamic emerging between the higher-level processes and the more basic sub-units.



System Dynamics



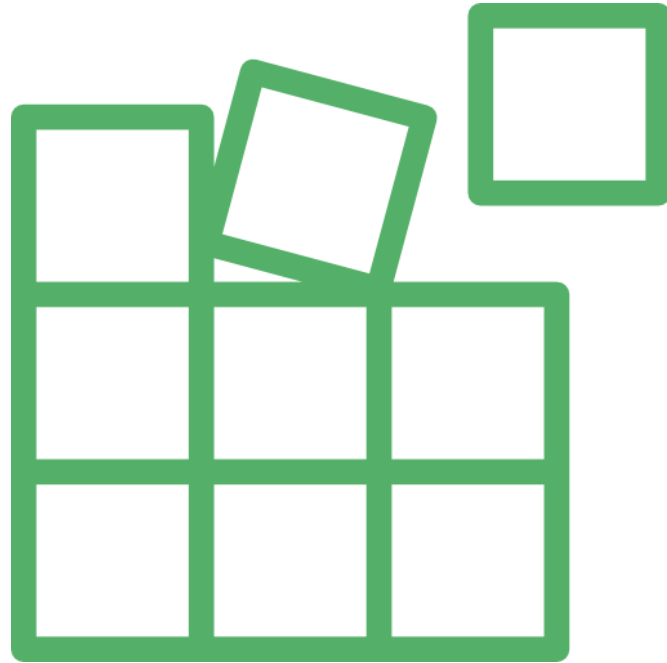
Overview

System dynamics is a branch of systems theory that tries to **model and understand the dynamic behavior of complex systems**. It deals with internal feedback loops and time delays that affect the behavior of the entire system. Stocks and flow diagrams are the foundation of system dynamics modeling as they let us look at and analyze the dynamic flows taking place within a system; whether this is information flows, the movement of products through a supply chain, or the flow of money through a financial system.

The key elements of system dynamics are **feedback loops, stocks and flows**. Systems thinking looks for the interplay between elements, that is the feedback loops through which elements are interconnected in effecting a joint outcome.

By modeling the interactions and thus visualizing the feedback loops, a causal loop diagram reveals the structure of a system. By understanding not only the structure to these relations but also the nature of those relations **it becomes possible to model and simulate a system's behavior over a certain time period**.





Stocks

A system stock is a store that can be quantified, e.g. a pile of sand or data in a database, the books in a bookstore. As a principle stocks generally change slowly, even when the flows into or out of them change suddenly. Therefore, stocks act as delays or buffers in systems.



Flows

Flows are entities that make stocks increase or decrease, like a tap or drain affects the level of water in a bathtub. For example, an underground gold deposit is a stock, out of which comes a flow of ore through mining. Flows are filling and draining, deposits and withdrawals, growth and decay.

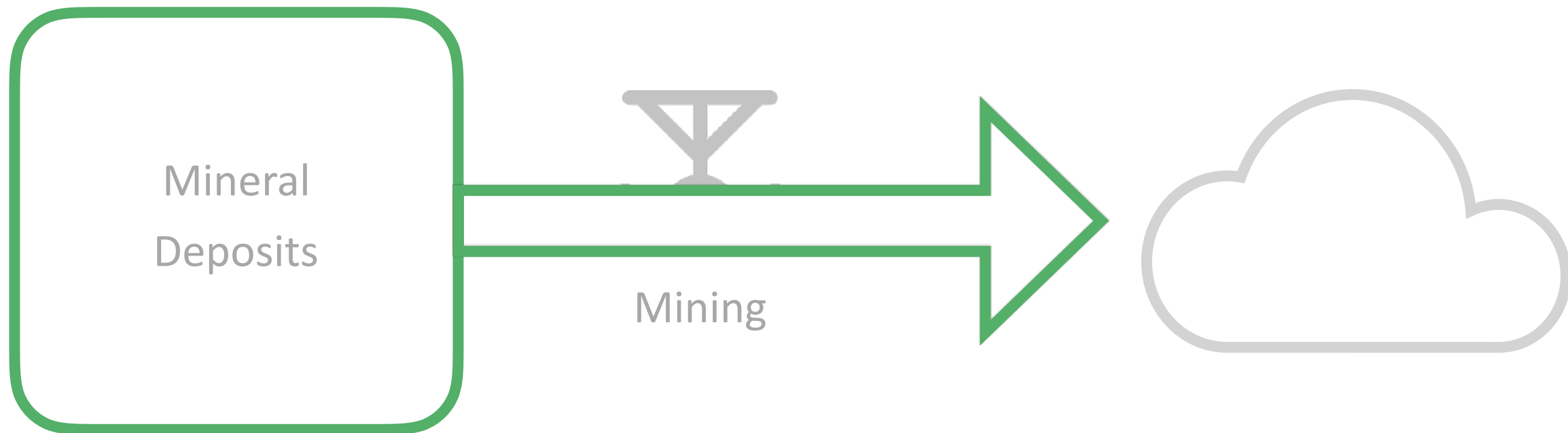
Stocks and Flows

Helps us to identify blockages and flows in the system



Illustration

Diagram of a stock of minerals depleted by mining



Stock vs Flow

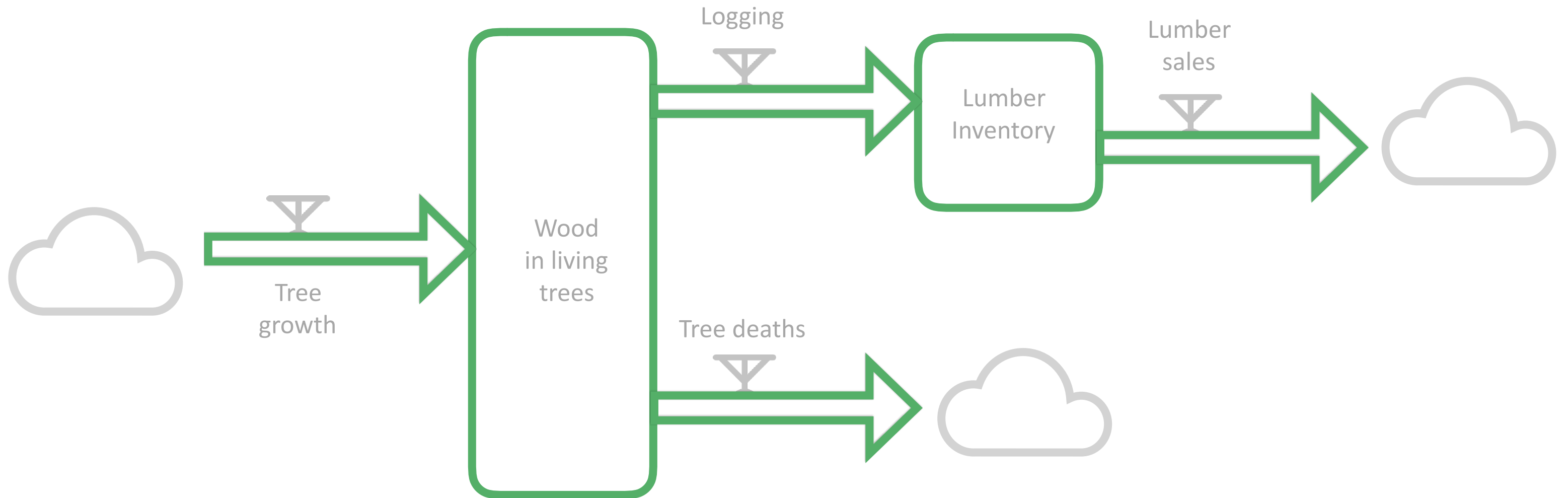
At first attempt it may not be too simple to distinguish stocks from flows. One way of distinguishing a flow from a stock is to ask what would happen in the system if time were to halt. Stocks, which are accumulations, would go on existing, while flows, in contrary, would disappear, because they are actions.

For example, at any given moment, the money in your savings account, the attendees at a sports game or the number of goods in a warehouse are measurable and, thus, are stocks; they would continue to exist if time were to stop. On the other hand, the bank's interest payment per day, the production rate of units per hour, and the movement of people in and out of a sports game, because these are flows and thus relative to time, they would no longer exist.



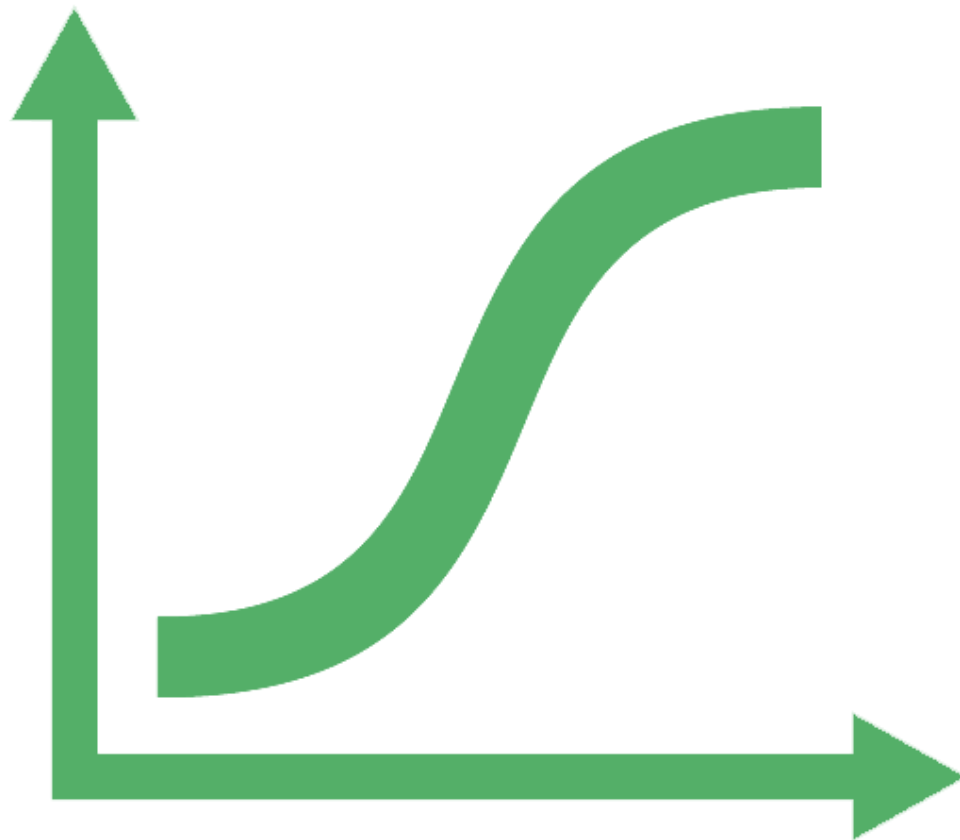
Illustration

A stock of lumber linked to a stock of trees in a forest



Behavior Over Time

Systems thinkers use graphs of system behavior to understand trends over time, rather than focusing attention on individual events. People also use behavior over time graphs to learn whether the system is approaching a goal or a limit, and if so, how soon or far away.



A behavior over time graph is a line diagram that shows how something has changed over a period of time. It creates a visual representation of what is happening in a system. Time is shown along the horizontal axis and the sizes of the variables are shown in the vertical dimension.

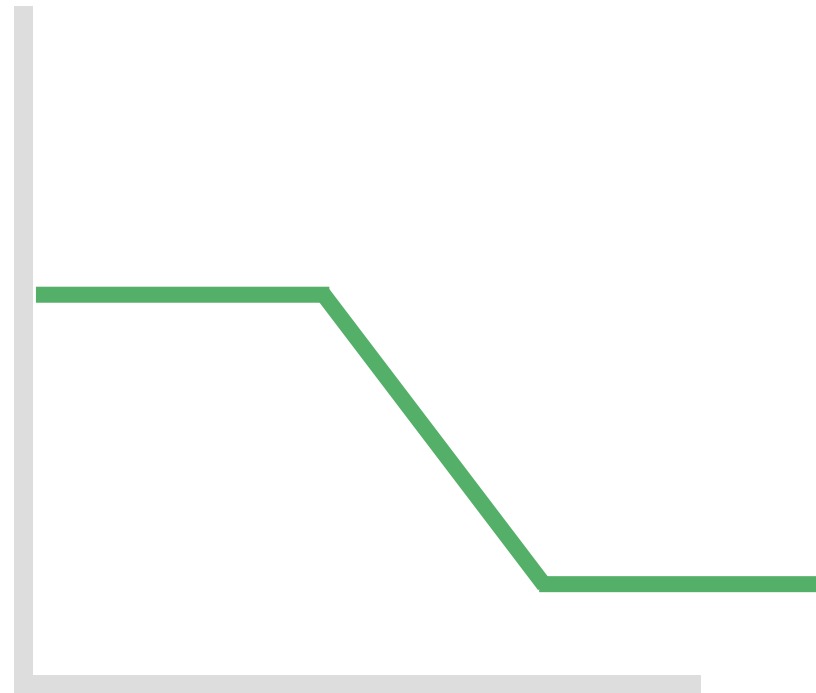
The aim is typically to gain a qualitative impression of the changes taking place rather than a detailed quantitative analysis. The pattern, captured by the form of the variable line, is important, as are the points at which that line changes shape or direction, while the precise numbers on the axes are often less important.

Principles



Rising

As long as the sum of all inflows exceeds the sum of all outflows, the level of the stock will rise.



Falling

As long as the sum of all outflows exceeds the sum of all inflows, the level of the stock will fall.



Stable

If the sum of all outflows equals the sum of all inflows, the stock level will not change; it will be held in dynamic equilibrium.

Stock and Flow Delays



The presence of stocks allows inflows and outflows to be independent of each other and temporarily out of balance with each other, creating delays. The time lags that come from slowly changing stocks can cause problems in systems, but they also can be sources of stability, thus both beneficial and detrimental. The time lags imposed by stocks allow room to maneuver, to experiment, and to revise policies that aren't working. Soil that has accumulated over centuries rarely erodes all at once, this gives us time to think of alternative agricultural practices before the critical point of failure.

Time lags often lead to "overshoot" which is the occurrence of a signal or function exceeding its target. For example, we might think of a town that has discover gold under its ground. When the gold is abundant, the population grows quickly. However, as the population grows it depletes the supply of gold. When gold becomes scarce, the population is stranded without a functioning economy and drops rapidly as people leave to search for work at other locations. Thus stock and flow delays can lead to crisis states due to overshoot.



Conclusion

Conclusion

Systems thinking is a holistic paradigm, which means that we are trying to understand the parts in relation to the whole. Thus in this guide to modeling systems, we have started from the most basic elementary parts and worked up to get to the system and its environment as a whole; the most complex level.

By interconnecting parts, we formed systems wherein the parts become interdependent in effecting each other and the state of the whole system they form part of. We looked at positive and negative synergies and how they can lead to emergence.

We then added a function to the system, where processes take in inputs and transform them into outputs. The system's output can be constructive - energy - or destructive to its environment - entropy. We then started to look at the system in relation to other systems and the overall environment; where the system boundary defines its autonomy and integrity.

The final section dealt with the dynamic behavior of a system within its environment and how feedback processes drive different patterns of development. We looked at positive and negative feedback as drivers of sustainable or unsustainable development and the systems control methods used to regulate these processes and maintain homeostasis.



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Reference & Resources

Lexico Dictionaries | English. (2020). System | Definition of System by Oxford Dictionary on Lexico.com also meaning of System. [online] Available at: <https://www.lexico.com/definition/system> [Accessed 1 Sep. 2020].

Universiteit Twente. (2016). Communication Studies theories: overview by category | University of Twente. [online] Available at: <https://www.utwente.nl/en/bms/communication-theories/> [Accessed 1 Sep. 2020].

Bagaria, J. (2014). Set Theory (Stanford Encyclopedia of Philosophy). [online] Stanford.edu. Available at: <https://plato.stanford.edu/entries/set-theory/#SetTheFouMat> [Accessed 1 Sep. 2020].

Wikiwand. (2010). Emergence | Wikiwand. [online] Available at: <https://www.wikiwand.com/en/Emergence> [Accessed 1 Sep. 2020].

Mathinsight.org. (2020). Function definition - Math Insight. [online] Available at: <https://mathinsight.org/definition/function#:~:text=A%20technical%20definition%20of%20a,related%20to%20exactly%20one%20output.&text=We%20can%20write%20the%20statement,notation%20f%3AX%E2%86%92Y>. [Accessed 1 Sep. 2020].

Wikiwand. (2020). System | Wikiwand. [online] Available at: <https://www.wikiwand.com/en/System> [Accessed 1 Sep. 2020].

How Systems Use Inputs & Outputs Video (2020). How Systems Use Inputs & Outputs | Study.com. [online] Study.com. Available at: <https://study.com/academy/lesson/how-systems-use-inputs-outputs.html> [Accessed 1 Sep. 2020].

Everton, H. (2017). System vs Process: What's the difference between them? [online] Get Organized - Online Calendar - Planner - CRM. Available at: <https://www.planplusonline.com/system-vs-process/#:~:text=What's%20a%20process%3F,together%20to%20make%20it%20function.&text=It's%20important%20that%20processes%20are,to%20produce%20a%20particular%20result>. [Accessed 1 Sep. 2020].

Wikiwand. (2020). Black box | Wikiwand. [online] Available at: https://www.wikiwand.com/en/Black_box [Accessed 1 Sep. 2020].

Mit.edu. (2020). Energy and Entropy. [online] Available at: <https://web.mit.edu/8.334/www/grades/projects/projects10/AlexanderPapageorge/Page4a.html#:~:text=Entropy%20is%20a%20measure%20of,a%20system%20can%20be%20configured.&text=As%20the%20temperature%20of%20the,fall%20into%20an%20ordered%20state>. [Accessed 1 Sep. 2020].

Wikiwand. (2020). Entropy (information theory) | Wikiwand. [online] Available at: [https://www.wikiwand.com/en/Entropy_\(information_theory\)](https://www.wikiwand.com/en/Entropy_(information_theory)) [Accessed 1 Sep. 2020].

Wikiwand. (2020). Potential energy | Wikiwand. [online] Available at: https://www.wikiwand.com/en/Potential_energy [Accessed 1 Sep. 2020].

Wikiwand. (2020). Dissipation | Wikiwand. [online] Available at: <https://www.wikiwand.com/en/Dissipation> [Accessed 1 Sep. 2020]

Varzi, A. (2013). Boundary (Stanford Encyclopedia of Philosophy). [online] Stanford.edu. Available at: <https://plato.stanford.edu/entries/boundary/> [Accessed 1 Sep. 2020].

Wikiwand. (2020). Open and closed systems in social science | Wikiwand. [online] Available at: https://www.wikiwand.com/en/Open_and_closed_systems_in_social_science#:~:text=An%20open%20system%20is%20defined,be%20isolated%20from%20their%20environment. [Accessed 1 Sep. 2020].

Wikiwand. (2020). Environment (systems) | Wikiwand. [online] Available at: [https://www.wikiwand.com/en/Environment_\(systems\)](https://www.wikiwand.com/en/Environment_(systems)) [Accessed 1 Sep. 2020].

OpenLearn. (2013). Managing complexity: A systems approach – introduction. [online] Available at: <https://www.open.edu/openlearn/science-maths-technology/computing-and-ict/systems-computer/managing-complexity-systems-approach-introduction/content-section-15.9.2> [Accessed 1 Sep. 2020].

The Systems Thinker. (2016). Causal Loop Construction: The Basics - The Systems Thinker. [online] Available at: <https://thesystemsthinker.com/causal-loop-construction-the-basics/> [Accessed 1 Sep. 2020].

BD Editors (2016). Positive Feedback. [online] Biology Dictionary. Available at: <https://biologydictionary.net/positive-feedback/#:~:text=Positive%20feedback%20is%20a%20process,action%20from%20continuing%20to%20occur>. [Accessed 1 Sep. 2020].

Wikiwand. (2020). Negative feedback | Wikiwand. [online] Available at: https://www.wikiwand.com/en/Negative_feedback [Accessed 1 Sep. 2020].

The Systems Thinker. (2016). Step-By-Step Stocks and Flows: Converting From Causal Loop Diagrams - The Systems Thinker. [online] Available at: <https://thesystemsthinker.com/step-by-step-stocks-and-flows-converting-from-causal-loop-diagrams/> [Accessed 1 Sep. 2020].

Contributors to Wikimedia projects (2005). Human Physiology/Homeostasis. [online] Wikibooks.org. Available at: https://en.wikibooks.org/wiki/Human_Physiology/Homeostasis [Accessed 1 Sep. 2020].

Wikiwand. (2020). Dissipative system | Wikiwand. [online] Available at: https://www.wikiwand.com/en/Dissipative_system#:~:text=A%20dissipative%20system%20is%20a,of%20as%20a%20dissipative%20system. [Accessed 1 Sep. 2020].

homeostasis | Definition, Examples, & Facts | Britannica. (2020). In: Encyclopædia Britannica. [online] Available at: <https://www.britannica.com/science/homeostasis> [Accessed 1 Sep. 2020].

Sciencedirect.com. (2014). Cybernetics - an overview | ScienceDirect Topics. [online] Available at: <https://www.sciencedirect.com/topics/computer-science/cybernetics> [Accessed 1 Sep. 2020].

Electrical4U (2012). Control System | Closed Loop Open Loop Control System | Electrical4U. [online] Electrical4U. Available at: <https://www.electrical4u.com/control-system-closed-loop-open-loop-control-system/#:~:text=A%20control%20system%20is%20a,system%2C%20which%20controls%20other%20systems.> [Accessed 1 Sep. 2020].

Tutorialspoint.com. (2020). Control Systems - Feedback - Tutorialspoint. [online] Available at: https://www.tutorialspoint.com/control_systems/control_systems_feedback.htm [Accessed 1 Sep. 2020].

Wikiwand. (2020). Adaptive system | Wikiwand. [online] Available at: https://www.wikiwand.com/en/Adaptive_system#:~:text=An%20adaptive%20system%20is%20a,or%20evolutionary%20adaptation%20in%20biology. [Accessed 1 Sep. 2020].

Wiener, N. (1961). Cybernetics, or control and communication in the animal and the machine (2nd ed.). [online] Available at: <http://www.allen-riley.com/utopia/cybernetics.pdf>

Delobelle, A. (1975). Feedback, Cybernetics and Sociology. Diogenes, [online] 23(91), pp.70–105. Available at: <https://journals.sagepub.com/doi/abs/10.1177/039219217502309105?journalCode=dioa> [Accessed 1 Sep. 2020].

Contributors to Wikimedia projects (2005). Human Physiology/Homeostasis. [online] Wikibooks.org. Available at: https://en.wikibooks.org/wiki/Human_Physiology/Homeostasis [Accessed 1 Sep. 2020].

Wikiwand. (2020). Actuator | Wikiwand. [online] Available at: <https://www.wikiwand.com/en/Actuator> [Accessed 1 Sep. 2020].

TheFreeDictionary.com. (2020). negative feedback loop. [online] Available at: <https://medical-dictionary.thefreedictionary.com/negative+feedback+loop> [Accessed 1 Sep. 2020].

Wikiwand. (2020). Second-order cybernetics | Wikiwand. [online] Available at: https://www.wikiwand.com/en/Second-order_cybernetics [Accessed 1 Sep. 2020].